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Abstract

In recent years, diffusion models have evolved substantially, leading to specialized models trained for specific objectives, such as generating content with unique styles and identities. However, many tasks remain challenging to tackle in real-world settings, like dancing video generation, and training models for these tasks requires enormous computational resources and large-scale datasets. To address these constraints, various model-fusion strategies have been proposed to combine the strengths of different models. Although these methods have achieved a measure of success, they are often restricted to homogeneous architectures and uniform training paradigms, limiting their usefulness across diverse model types.

This thesis introduces **AdaptiveFuse**, a novel noise-fusion framework that seamlessly fuses models with different architectures, training methods, and sampling schedules without requiring further training. By leveraging the complementary capabilities of specialized models, AdaptiveFuse excels in complex, real-world applications. We also provide a comprehensive study on scheduling parameter selection for the fusion process, identifying the most effective strategies for optimal generation performance under varied conditions. Our proposed method is demonstrated in two main scenarios, such as character image stylization and dancing video generation, where it achieves higher visual fidelity and better identity preservation compared to existing approaches. By enabling the fusion of heterogeneous models, AdaptiveFuse significantly expands the potential of generative modeling in diverse areas such as video editing, inpainting, and novel view synthesis, paving the way for tackling more complex and resource-intensive tasks.

Keywords: Diffusion Models, Model Fusion, Video Generation, Identity Preservation.

摘要

近些年来，扩散模型（diffusion models）发展迅速，并衍生出了许多针对特定目标而训练的专用模型，例如生成独特风格或是保留特定身份特征的内容。然而，在许多实际应用场景（如舞蹈视频生成）依然存在诸多挑战。要训练能够胜任这些任务的模型，往往需要庞大的计算资源和海量数据。为了解决这些限制，研究者们提出了各种模型融合策略，旨在将不同模型的优势相结合。尽管这些方法在一定程度上取得了成功，但它们通常局限于同构的网络架构和统一的训练范式，使其在异质性较高的模型环境中难以推广。

本论文提出了一种名为 **AdaptiveFuse** 的全新噪声融合框架，可在无需额外训练的情况下，将具有不同架构、训练协议以及采样方式的模型无缝融合在一起。通过利用各专用模型的互补特性，AdaptiveFuse 在处理复杂的现实任务时表现出色。我们还对融合过程中各项调度参数的选择进行了系统研究，以确定在多种场景下获得最优生成性能的策略。论文中所述方法主要应用于角色图像风格化与舞蹈视频生成这两个场景，并在视觉质量和身份保真度方面优于现有方法。通过融合具备多样化特性的模型，AdaptiveFuse 显著拓展了生成式建模在视频编辑、修复（inpainting）以及新视角合成等领域的潜能，为应对更复杂、更高需求的实际应用奠定了基础。

关键词：扩散模型、模型融合、视频生成、身份保真

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List of Symbols

Symbol	Description
VAE	Variational Autoencoder
GAN	Generative Adversarial Network
DDPM	Denoising Diffusion Probabilistic Models
DDIM	Denoising Diffusion Implicit Models
LDM	Latent Diffusion Models
LoRA	Low-Rank Adaptation (parameter-efficient fine-tuning)
ControlNet	Pose-based or condition-based guidance for diffusion
IP-Adapter	Adapter for multi-conditioned generation (image+text)
CLIP	Contrastive Language-Image Pre-training (text-image encoder)
FID	Fréchet Inception Distance (image quality metric)
FVD	Fréchet Video Distance (video consistency metric)
PSNR	Peak Signal-to-Noise Ratio (image fidelity metric)
SSIM	Structural Similarity Index Measure (image fidelity metric)
LPIPS	Learned Perceptual Image Patch Similarity (perceptual metric)
Adaface	Face identity metric based on cosine similarity



Chapter 1. Introduction

In this chapter, we provide an overview of the motivation, background, and significance of the proposed model fusion approach, reiterating how it addresses key challenges in multi-objective generative tasks. It begins by examining the technical evolution of generative modeling, highlighting the unique benefits and limitations of diffusion-based approaches. Subsequently, the chapter investigates the state of current research on diffusion models and model fusion, emphasizing the need for a more flexible, training-free solution. Finally, it delineates the core objectives and contributions of this thesis, situating the proposed framework within the broader diffusion model landscape, and concludes by outlining the thesis organization and offering a preview of the subsequent chapters.

1.1 Research Background and Significance

Generative models have progressed rapidly, showcasing a remarkable capacity for generating increasingly realistic data across diverse domains. Early foundational methods, like Variational Autoencoders (VAEs) [1], Generative Adversarial Networks (GANs) [2], and Normalizing Flows [3], laid the groundwork for modern generative frameworks capable of tasks ranging from high-quality image production to multimodal content creation. By mapping latent distributions onto high-dimensional data manifolds, these models have unlocked compelling frontiers in controllable generation, fidelity, and distribution coverage.

Within this evolving sphere of generative approaches, diffusion models have recently emerged as a major breakthrough, surpassing traditional techniques in numerous applications [4–7]. Their underlying principle revolves around a forward-backward stochastic process: a forward or “noising” phase that progressively corrupts a clean data distribution until it becomes noise-dominated, followed by a learned backward denoising phase. This iterative method starkly contrasts the single-shot strategy employed by GANs, thereby reducing mode collapse and stabilizing training. Moreover, variants such as score-based models [6, 8], denoising diffusion probabilistic models (DDPMs) [5], and latent diffusion models (LDMs) [7] have improved dif-



fusion theory, making generation more scalable and efficient for high-resolution data.

Beyond their aptitude for producing visually coherent images, diffusion models also excel in practical domains, including image synthesis [9, 10] and video generation [11, 12]. By repeatedly denoising random noise, these methods develop robustness and controllability, enabling applications in areas such as style transfer [13], identity preservation [14], and motion generation [15, 16]. Nonetheless, diffusion models frequently struggle when multiple specialized objectives must be combined into a single process [17]. For example, synthesizing a dancing video that is realistic in human motion yet faithful to an individual’s identity requires specialized capacities that are typically hard to merge within a single model without large-scale re-training or suboptimal outcomes [18, 19].

Model fusion techniques have therefore emerged as a pragmatic approach, aiming to harness the strengths of existing trained models without incurring the burden of comprehensive re-training [17]. However, common fusion strategies usually restrict themselves to networks sharing compatible architectures or training paradigms, thus limiting their utility across more diverse model sets [18, 19]. Consequently, real-world applications that demand multiple specialized generative functionalities lack a sufficiently flexible fusion mechanism.

This thesis addresses this limitation by presenting **AdaptiveFuse**, a noise-level fusion framework that seamlessly merges diffusion models with differing architectures, training methods, and sampling schedules. Furthermore, this approach adaptively fuses each model’s noise residual predictions in each denoising step, thereby overcoming architectural constraints and large-scale re-training. Without requiring any additional training, AdaptiveFuse enables the fusion of specialized models to perform multifaceted generation tasks, such as dancing video creation and character image stylization, thereby surpassing the deficiencies of earlier fusion methods. By proposing a noise-based integration strategy, AdaptiveFuse broadens the scope of diffusion modeling, facilitating the creation of more versatile and powerful generative pipelines.

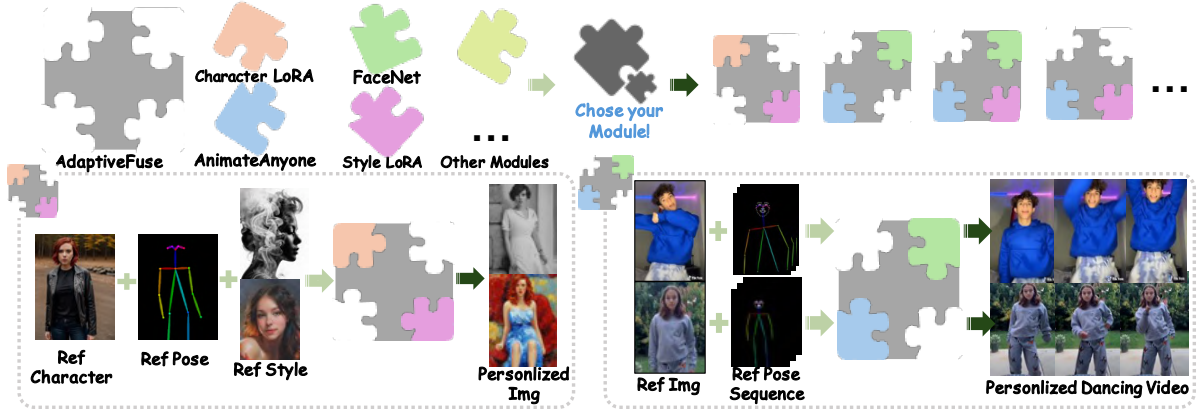


Figure 1.1 We proposed AdaptiveFuse, a model-agnostic noise fusion approach for seamlessly integrating multiple pre-trained diffusion models, enabling complex real-world applications such as high-quality conditioned image generation and dancing video generation.

1.2 Related Work

In recent years, diffusion-based models have established themselves as powerful frameworks for high-fidelity image, video, and multimodal generation. As thoroughly reviewed in Chapter 2, these approaches evolved from earlier generative paradigms such as Variational Autoencoders (VAEs) [1], Generative Adversarial Networks (GANs) [2], and Normalizing Flows [3], each of which laid important groundwork but also faced notable challenges (e.g., training instability or limited scalability) [20].

1.2.1 Fundamentals of Diffusion Models

The underlying mechanism of diffusion models involves gradually corrupting data in a forward process and learning to invert this corruption step-by-step via a reverse diffusion procedure [4–6]. Their multi-step denoising paradigm effectively addresses issues like mode collapse that often hinder GANs, thereby enabling more stable training and improved coverage of the data distribution. As discussed in Section 2.2 of the *Literature Review*, foundational work in Denoising Diffusion Probabilistic Models (DDPMs) and Denoising Diffusion Implicit Models (DDIMs) [21, 22] has since been extended to latent diffusion [7], resulting in significantly reduced computational overhead.



1.2.2 Diffusion Models for Image Generation

Recent diffusion-based methods, such as LDMs [7], stable diffusion frameworks [9, 10], and personalization approaches (e.g., Low-Rank Adaptation (LoRA)-based fine-tuning [23]), have pushed the boundaries of image generation quality, controllability, and efficiency. As summarized in Section 2.2 and Section 2.3, these innovations facilitate complex tasks like style transfer [13] or identity preservation [14] through specialized modules (e.g., ControlNet [24] and IP-Adapter [25]).

1.2.3 Diffusion Models for Video Generation

Extending diffusion approaches to videos entails managing spatiotemporal consistency and realistic motion. Prior research employs specialized attention mechanisms [11, 12] or separate motion modules (e.g., AnimateDiff [12]) to generate coherent multi-frame sequences. As discussed in Section 2.5, maintaining identity consistency, especially in human-centric or dance-oriented videos, remains an active research area where minor inconsistencies can compound over time [16, 26].

1.2.4 Model Fusion Techniques

While diffusion models excel at specific tasks, real-world applications often demand multiple specialized capabilities in a single generative pipeline. Model fusion addresses this need by combining pre-trained models to leverage their respective strengths. Existing methods include:

- **Weight-based Fusion:** Methods like ZipLoRA [18] merge LoRA weight updates from different specializations (e.g., style, subject), although they typically require shared base architectures.
- **Switch-based Fusion:** LoRA Composite [19] swaps LoRA modules at inference, enabling multiple styles or subjects, but may lack fine-grained blending.
- **Noise-based Fusion:** Approaches such as NVS-Solver [17] and Time Reversal Fusion [27]



manipulate noise residuals to reconcile multiple diffusion models, however, often assuming homogeneous architectures or sampling schemes.

As highlighted in Section 2.6 of the *Literature Review*, each fusion methods has inherent limitations regarding flexibility, architectural compatibility, or re-training overhead.

1.2.5 Summary of Related Work

While existing diffusion models demonstrate remarkable performance individually, fusing them to handle diverse objectives (e.g., realistic motion, unique identities, stylized outputs) poses significant challenges. The literature underscores the trade-off between specialization and flexibility: most methods either require shared architectures or re-training, limiting their adaptability across heterogeneous model sets. This thesis tackles these gaps by proposing **AdaptiveFuse**, a training-free noise-level fusion framework that merges diffusion models with different designs and training conditions. By dynamically integrating noise residuals, we aim to provide a unified pipeline that combines specialized skills in tasks such as dancing video generation and personalized image stylization.

1.3 Contributions and Significance

In this thesis, we propose a noise-based fusion method, **AdaptiveFuse**, targeting obstacles in combining diverse diffusion models. Our principal contributions include:

1. **AdaptiveFuse Framework:** We introduce a noise-focused integration mechanism for training-free fusion among various diffusion models, removing architecture and training dependencies. By adjusting noise strengths at each denoising step, we can dynamically fuse specialized models.
2. **Comparative Fusion Strategies:** We investigate switch-based, half-weight, noise-based, and manual approaches, providing a comparative analysis that highlights trade-offs and optimal choices under varied conditions.



3. **Applications to Video Generation and Image Stylization:** We demonstrate AdaptiveFuse in dancing video creation and character image stylization, surpassing baselines in preserving temporal consistency, fidelity, and identity.

By providing a flexible, robust noise fusion paradigm, our work significantly advances the adaptability of diffusion-based generative modeling, rendering it suited for scenarios where multiple specialized skills must coexist in a single generative pipeline.

1.4 Thesis Outline

The central theme throughout is the development of a flexible, training-free noise-level fusion method, **AdaptiveFuse**, for merging multiple diffusion models in complex generative tasks.

- **Chapter 1 – Introduction** Presents the motivation, significance, and objectives of this research. It also introduces the central challenges in multi-objective generative modeling and clarifies how diffusion-based techniques can address them.
- **Chapter 2 – Literature Review** Reviews foundational and recent work on diffusion models, including their application to image and video generation, personalization, and multi-conditioned tasks. It also covers existing model fusion techniques and highlights the gaps that AdaptiveFuse aims to fill.
- **Chapter 3 – Character LoRA and Style LoRA Fusion** Proposes a noise-based fusion strategy for integrating multiple LoRA modules. The chapter explains how the framework merges distinct LoRAs (e.g., for identity and for style) within a single diffusion pipeline, supported by experimental analysis.
- **Chapter 4 – Dancing Video Generation Fusion** Explores an advanced application of AdaptiveFuse to dancing video generation. It demonstrates how specialized models (e.g., for facial detail and motion dynamics) can be adaptively combined, discusses the methodology and experimental setup, and highlights improvements over existing methods.



- **Chapter 5 – Conclusion and Future Work** Summarizes the contributions of this thesis and underscores its limitations. Finally, it presents prospective research directions aimed at further enhancing noise-oriented diffusion model fusion.





Chapter 2. Literature Review

This chapter provides an extensive review of diffusion models and their applications in image and video generation, personalization, and multi-conditioned tasks. It begins with a discussion of foundational concepts and state-of-the-art architectures, including DDPMs, DDIMs, and the Stable Diffusion framework. The review then explores advanced strategies for improving sampling efficiency, integrating control signals, and personalizing outputs to specific individuals or styles. It also addresses the challenges and limitations of current model fusion techniques, setting the stage for the proposed AdaptiveFuse approach introduced in subsequent chapters.

2.1 Historical Background on Generative Models

Generative modeling has been central to machine learning research, encompassing methods such as VAEs, GANs, and normalizing flows [4–6, 9]. While GANs gained popularity due to their high-quality image outputs, they often suffer from unstable training dynamics and mode collapse, which can limit their scalability [2]. VAEs offer more stable training but can yield blurrier images due to their latent-variable formulation [1]. Normalizing flows require maintaining tractable Jacobians, a constraint that grows more complex as models scale [3].

Diffusion models emerged to address these limitations by systematically breaking down the image generation process into a series of denoising steps. This approach effectively separates generation into simpler, iterative procedures that can be more robust and yield high-fidelity results [7, 28, 29]. Compared to GANs or VAEs, diffusion models frequently demonstrate superior control over the sampling trajectory, enabling advanced editing, stylization, and personalization.

2.2 Diffusion Models

For a comprehensive overview of diffusion-based generative modeling [29], diffusion models represent a powerful class of generative models that transform simple noise distributions into complex data distributions through a two-step iterative process: a *forward process* that gradually adds noise to data, and a *reverse process* that learns to remove noise step-by-step. This paradigm has demonstrated exceptional performance in high-quality image, audio, and video synthesis, marking a significant milestone in generative modeling research [4–6]. Beyond these foundational works, theoretical and algorithmic advances have addressed sampling efficiency, scoring functions, and model interpretability [29–31].

2.2.1 Denoising Diffusion Probabilistic Models (DDPM)

DDPMs[21, 22, 32] serve as a foundational framework in this area. Consider a data sample x_0 drawn from a real data distribution $q(x_0)$. The *forward diffusion process* incrementally adds Gaussian noise to produce a sequence $\{x_t\}_{t=1}^T$:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I}) \quad (2-1)$$

Here, $\{\beta_t\}_{t=1}^T$ is a noise schedule, and as $t \rightarrow T$, x_T approaches an isotropic Gaussian distribution. One can express x_t directly in terms of x_0 :

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \text{where } \alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{i=1}^t \alpha_i, \text{ and } \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (2-2)$$

As shown in Figure 2.1, this forward process progressively destroys the data structure, turning a clean sample into nearly pure noise. Alternative parameterizations and improved noise schedules have been explored to enhance convergence and sample quality [21, 32].

The *reverse diffusion process* aims to reconstruct x_0 from x_T by learning a parameterized distribution $p_\theta(x_{t-1}|x_t)$. While the true reverse distribution is intractable, one can train a neural network, typically a U-Net, to predict the added noise ϵ at each step:

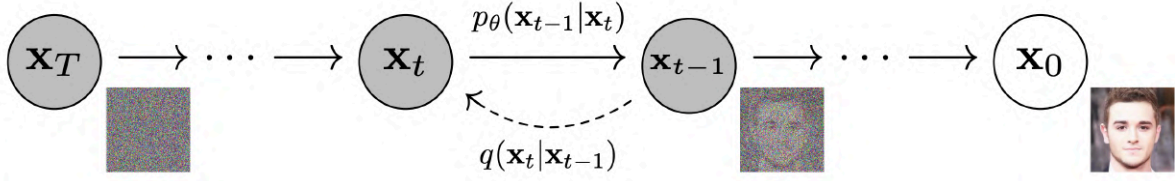


Figure 2.1 Illustration of the process in DDPM.

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t)). \quad (2-3)$$

In practice, Σ_{θ} is often fixed or learned to simplify training. The network is trained by minimizing the following loss function, which measures the difference between the predicted noise $\epsilon_{\theta}(x_t, t)$ and the true noise ϵ :

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]. \quad (2-4)$$

Various loss reweighting and architectural refinements have also been studied to improve training stability and model expressiveness [28, 33].

2.2.2 Training and Sampling

Training involves sampling (x_0, t, ϵ) , computing x_t , and updating θ to minimize the noise prediction error. Once trained, sampling (generation) starts from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and applies the learned reverse steps to gradually denoise:

$$x_{t-1} = \mu_{\theta}(x_t, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, \mathbf{I}),$$

iterating until x_0 is obtained. Advanced samplers and optimization strategies focus on reducing the number of inference steps while maintaining fidelity [31, 34].

2.2.3 Denoising Diffusion Implicit Models (DDIM)

DDIMs [22] provide a deterministic or more flexible sampling scheme by introducing a parameter η . When $\eta = 0$, the process becomes a deterministic mapping, accelerating sampling



and enabling easier manipulation such as DDIM inversion. Exploiting DDIM inversion, techniques like Prompt-to-Prompt editing [35, 36] enable sophisticated image editing directly in the generative process. The DDIM reverse process is defined as:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}x_0 + \sqrt{1 - \bar{\alpha}_{t-1}}\epsilon_t \quad (2-5)$$

with adjusted parameters ensuring a consistent reverse mapping. This allows faster and more stable sampling and provides a pathway to advanced editing techniques by navigating in the latent (or intermediate) spaces of the model.

2.3 Stable Diffusion Model

While DDPMs and DDIMs produce impressive results in pixel space, they can be computationally expensive, especially at high resolutions. The Stable Diffusion model [7] addresses this challenge by performing the diffusion process in a *lower-dimensional latent space*, thereby reducing computational overhead without sacrificing output fidelity. Other latent diffusion frameworks and improvements are discussed in [37, 38].

Variational Autoencoder (VAE)

A key component of Stable Diffusion is a VAE (E, D) . The encoder E compresses an image $x \in \mathbb{R}^{H \times W \times 3}$ into a latent code $z = E(x)$, and the decoder D reconstructs an image $\hat{x} = D(z)$. By operating on the latent code z instead of the original high-resolution image, the model circumvents the high dimensionality of pixel space.

Latent Diffusion Process

Stable Diffusion (SD) learns to generate images by iteratively denoising a Gaussian noise vector z_T into a meaningful representation z_0 over T timesteps. Specifically, the diffusion process occurs on the latent code z , rather than on raw pixel data:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i,$$

where α_t is a schedule hyperparameter and $\epsilon \sim \mathcal{N}(0, I)$. The forward process gradually corrupts z_0 with noise until it becomes nearly random at step T . During training, a U-Net ϵ_θ is trained to predict the added noise at each timestep t . The training objective,

$$\mathcal{L} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon_\theta(z_t, t, c) \right\|_2^2 \right], \quad (2-6)$$

minimizes the discrepancy between predicted and actual noise, where c is a conditioning vector (e.g., from a CLIP text encoder [39]).

Inference (Reverse Diffusion): At inference, the model starts from a random Gaussian latent z_T and iteratively denoises it by applying the learned reverse diffusion steps. After T iterations, the final latent code z_0 is obtained, which is then decoded into an output image $\hat{x} = D(z_0)$. This latent-space denoising framework brings significant efficiency gains over pixel-space diffusion, particularly at higher resolutions.

U-Net with Cross-Attention for Conditional Guidance

The core denoising step in Stable Diffusion uses a U-Net architecture, selected for its efficacy in multi-scale feature extraction and reconstruction. As shown in Figure 2.2, the U-Net's symmetrical design with skip connections helps preserve both local details and global context, ensuring fine-grained and coherent generation. Architectural and sampling refinements for U-Net-based diffusion models appear in [30, 31].

To enable text-conditioning, Stable Diffusion incorporates a pre-trained text encoder (e.g., CLIP) and integrates its embeddings into the U-Net via cross-attention layers. Let y be a text prompt and $\tau(y)$ its embedding. During denoising, cross-attention aligns image features $\phi_i(z_t)$ with textual representations $\tau(y)$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \quad (2-7)$$

where

$$Q = W_Q \phi_i(z_t), \quad K = W_K \tau(y), \quad V = W_V \tau(y),$$

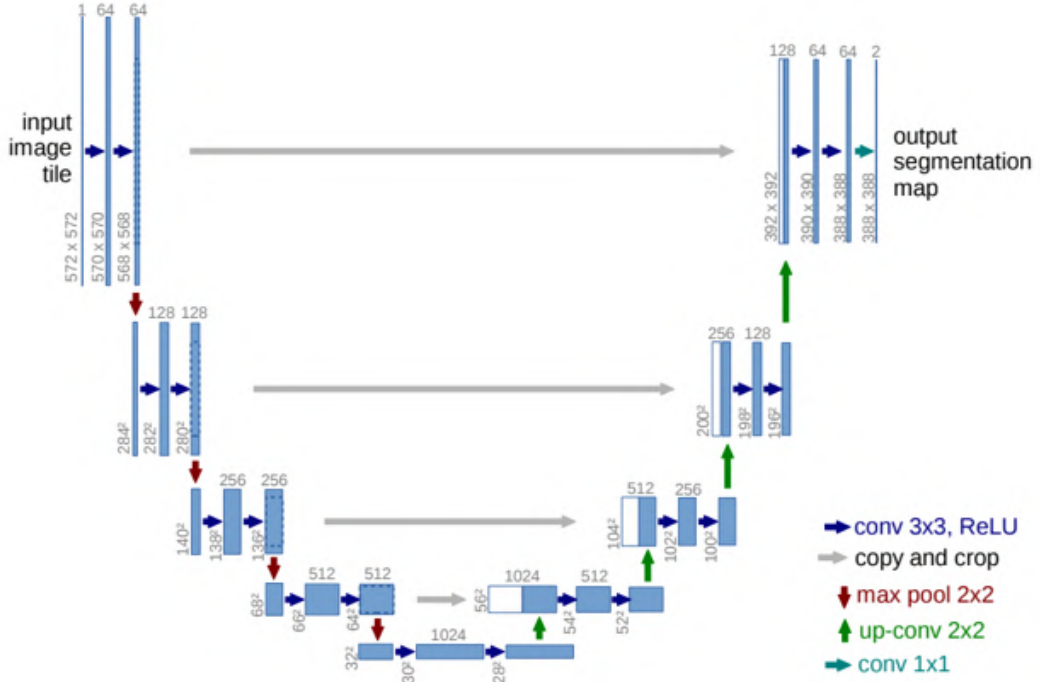


Figure 2.2 U-Net architecture used in Stable Diffusion. Skip connections between encoder and decoder layers preserve spatial detail and aid in high-quality image reconstruction.

and W_Q , W_K , W_V are learnable projection matrices. This mechanism infuses semantic guidance into the diffusion steps, enabling flexible text-driven or multimodal generation. Refinements to prompt conditioning and multimodal guidance have been explored in [40, 41].

Summary

In summary, Stable Diffusion leverages an autoencoder to compress images into a latent code, mitigating the computational burden of pixel-space diffusion. A U-Net operating on these latent representations performs iterative denoising, guided by cross-attention for flexible conditioning. This paradigm stands as a significant advancement over earlier approaches, enabling highly efficient and controllable image synthesis. It provides a cornerstone for subsequent developments that extend diffusion models to more specialized tasks, such as personalized stylization, inpainting, and high-resolution video generation.

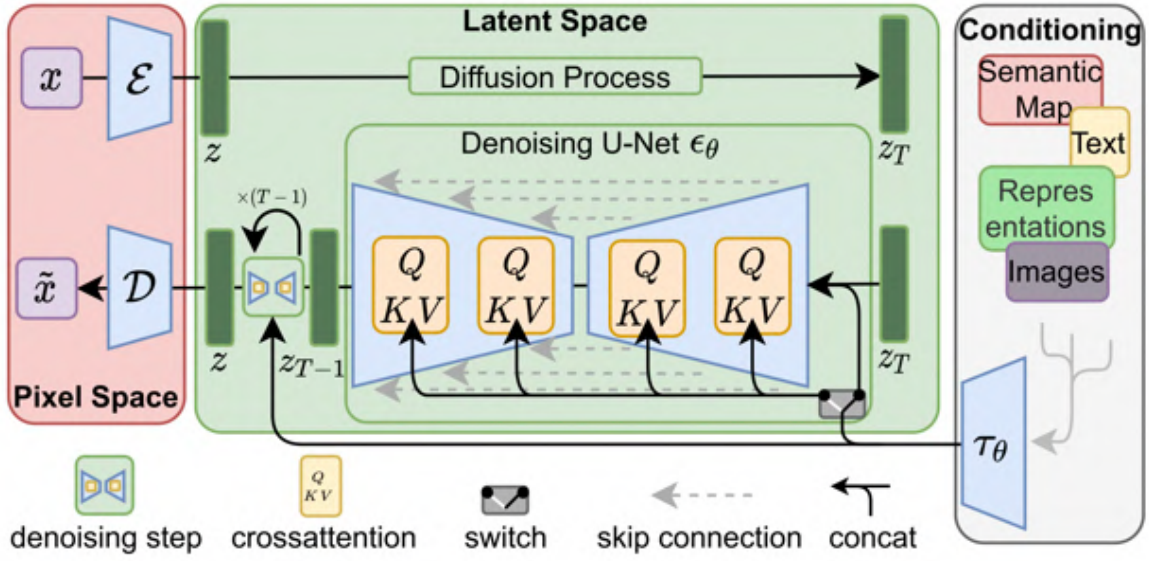


Figure 2.3 Stable Diffusion pipeline: The process begins with a text prompt, encoded by a text encoder. The latent diffusion process occurs in the compressed latent space, guided by text-image cross-attention. Finally, the VAE decoder reconstructs the final high-resolution image.

2.4 Diffusion Models for Image Generation

Diffusion models have achieved state-of-the-art performance in image generation tasks, surpassing traditional GAN-based approaches in terms of image quality and diversity. The versatility of diffusion models has led to their adoption in a wide range of applications, such as image generation [4–9, 32, 42], image super-resolution [7, 28, 32, 43–45], image inpainting [4, 6–8, 45, 46], image-to-image translation [47] and image editing [48–50] and conditioned image generation.

The conditioned image generation has pushed performance further on complex datasets [47, 51]. Key advancements include:

Latent Diffusion Models (LDMs): LDMs [7] leverage the latent space of autoencoders, reducing computational requirements and enabling high-resolution image synthesis.

Low-Rank Adaptation (LoRAs): LoRA [23] is a technique that fine-tunes large models more efficiently by injecting low-rank weight matrices into their layers. Originally introduced in the context of language models, LoRA has also become a popular approach for personalizing or adapting large-scale diffusion models without re-training them entirely. By focusing on only a fraction of the parameters, LoRA can add new styles, identities, or concepts to a diffusion

model at reduced computational and memory costs.

ControlNet: ControlNet [24] adds a parallel “trainable copy” of Stable Diffusion’s encoder – decoder blocks to process auxiliary inputs (e.g., edges, segmentation maps, pose heatmaps). As shown in Figure 2.4, these copied modules run alongside the original U-Net and are merged via *zero convolution* layers, which initialize to zero weights so that the base model remains unchanged at first. During training, the model learns how much of the ControlNet branch’s features to incorporate for fine-grained control. This design yields a flexible framework for integrating diverse spatial conditions without overwriting the powerful priors in the pre-trained Stable Diffusion weights, and has inspired further work on multimodal guidance and complex conditioning [52, 53].

IP-Adapter: IP-Adapter [25] fuses image and text prompts for personalized, multi-conditioned generation. As shown in Figure 2.6, a frozen image encoder extracts reference-image features, which are then projected into the diffusion U-Net. Meanwhile, a frozen text encoder processes text prompts. By *decoupling cross-attention*, IP-Adapter allows image-derived cues (e.g., style or subject identity) and text guidance (semantic constraints) to coexist without conflict. Similar methods [41] refine prompt conditioning, but IP-Adapter stands out for seamlessly integrating image-based cues into the diffusion process for more flexible personalization.

IP-Adapter FaceID: Building on the IP-Adapter and IP-Adapter FaceID aligns generated faces with reference identities [25]. Similar image-to-image translation techniques [47, 51] further enhance personalization.

Style Transfer and Identity Preservation: Techniques like StyleDrop [13] and Instant Identity [14] focus on transferring specific styles or preserving identities. Additional personalization approaches, such as DreamBooth [54] and Textual Inversion [55], learn new concepts from a handful of images, thereby expanding the model’s versatility.

2.5 Diffusion Models for Video Generation

Building upon LDMs for image generation [7, 9, 10], recent work introduces methods to animate text-to-image (T2I) pipelines. *AnimateDiff* [12] is one such framework, shown in

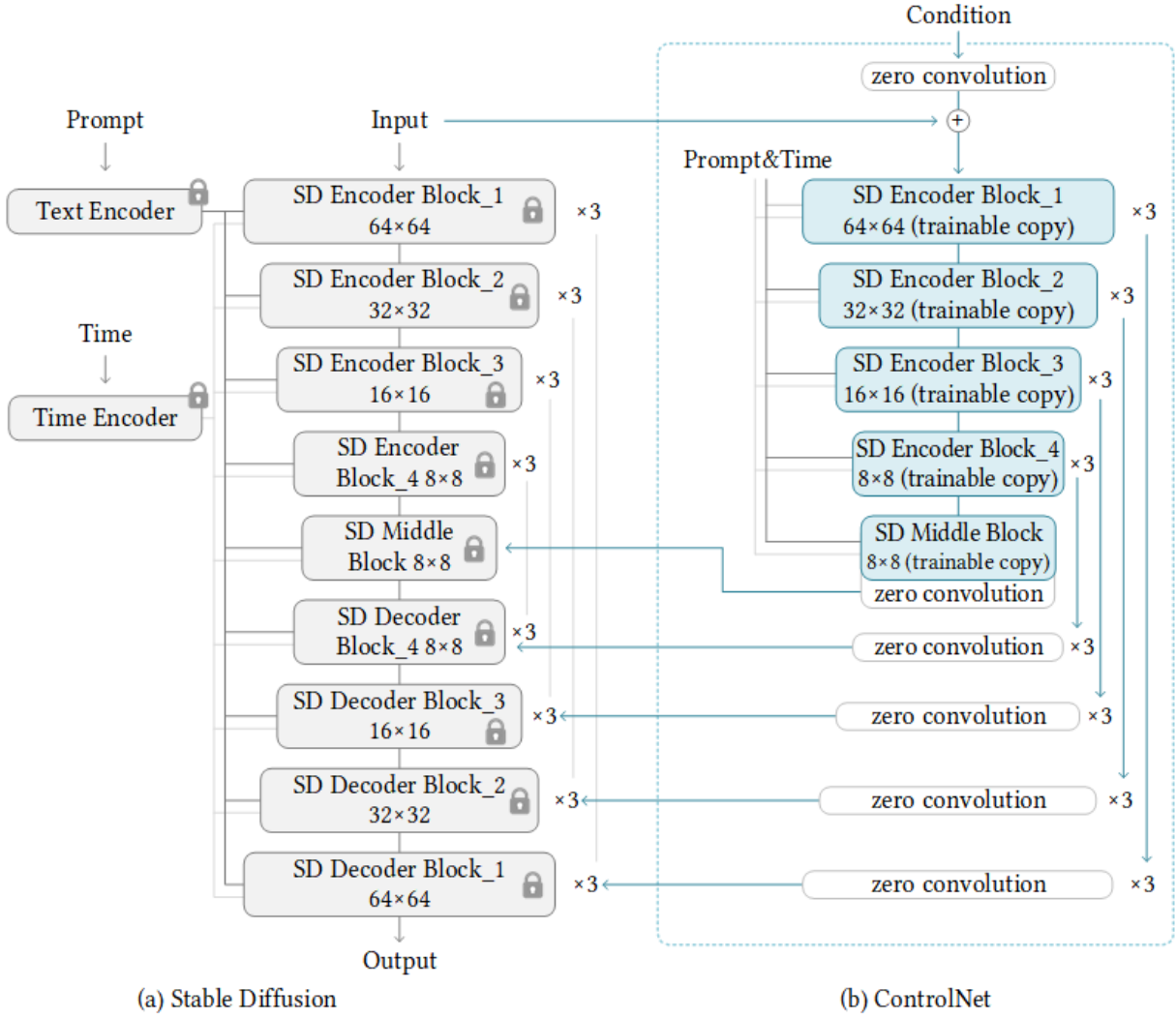


Figure 2.4 Overview of ControlNet. (a) Stable Diffusion layers remain frozen (gray locks). (b) A parallel trainable branch, equipped with zero-convolution blocks, injects conditional features (edges, poses, etc.) into the main U-Net.

Figure 2.6, where a *plug-and-play motion module* is inserted into a personalized T2I U-Net without re-training the original weights. This module, trained on real-world videos, injects motion priors that yield coherent animations aligned with the model’s unique style. These components allow AnimateDiff to generate smooth, visually consistent videos across diverse domains while preserving each T2I model’s specialized appearance or content knowledge.

Building on the motion module architecture, recent work has focused particularly on human-centric video generation, such as dance or pose-driven animation [16, 26, 56–59], where fine-grained details like facial expressions and body shape must remain consistent throughout the video. Despite the progress, preserving detailed facial features remains an open challenge; sub-

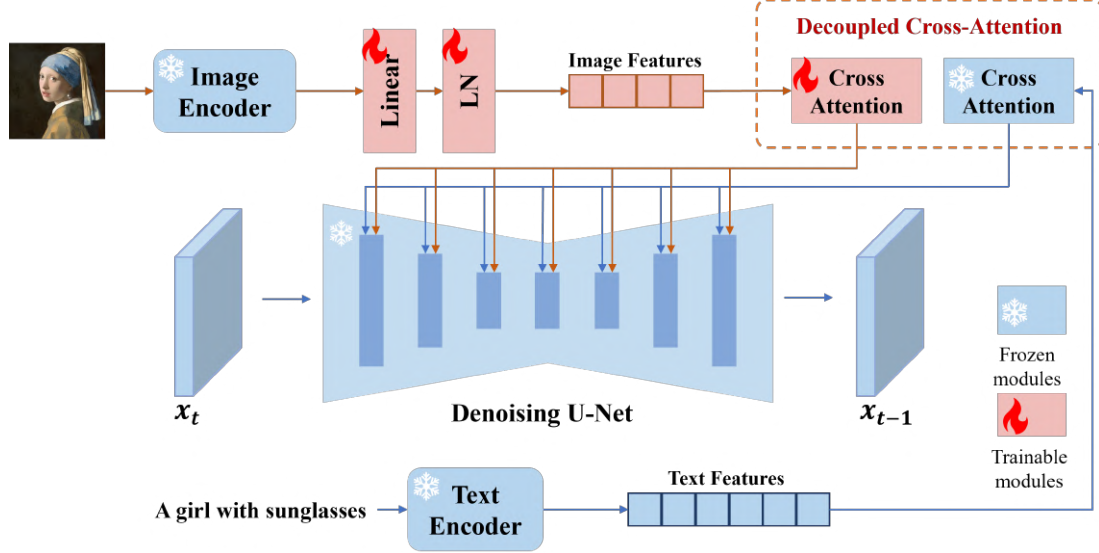


Figure 2.5 Overview of IP-Adapter. Frozen modules (blue) encode images and text, while trainable blocks (red) project image features into the U-Net, enabling decoupled cross-attention.

the deviations in early frames often compound over time, leading to flickering or drift. Ongoing efforts to address this issue include integrating stronger identity constraints [60] or leveraging disentangled pose-and-appearance representations [61, 62] to stabilize the generation process.

Overall, diffusion-based video generation has shown promise for complex, multi-frame tasks, yet further research is needed to handle longer sequences and ensure robust identity preservation, especially in demanding scenarios like dance and motion-intensive actions.

2.6 Model Fusion Approaches

Model fusion strategies aim to combine the strengths of multiple pre-trained models. Approaches that merge pre-trained diffusion models or integrate them with other frameworks, such as VAEs or GANs, appear in [63, 64].

Weight-based Fusion: ZipLoRA [18] merges LoRA weight updates (e.g., for a subject

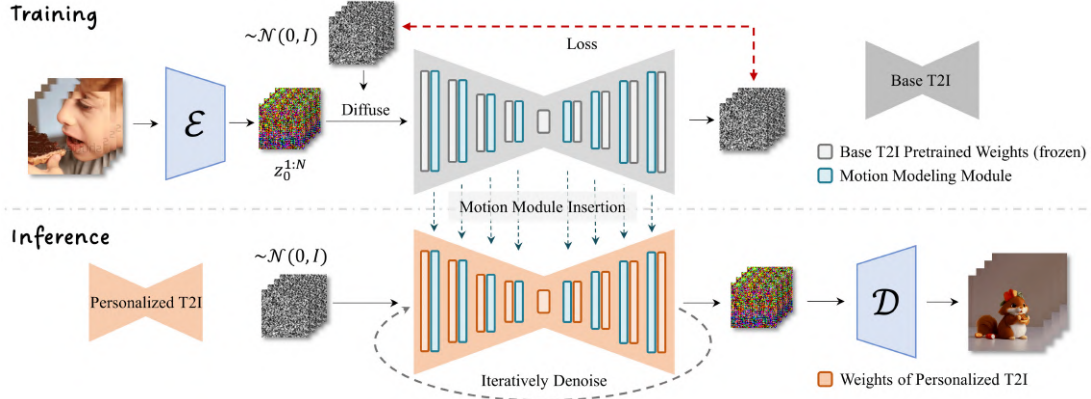


Figure 2.6 AnimateDiff inserts a single motion module (highlighted) into a pre-trained T2I model. This module infuses temporal coherence and allows optional fine-tuning (MotionLoRA) for new motion patterns.

and a style) by reducing column-wise overlaps in the LoRA matrices. It introduces a lightweight optimization step that learns “merger coefficients” to avoid destructive interference among columns that are highly aligned, ensuring each LoRA’s distinct knowledge is retained. While this approach effectively combines multiple LoRAs into a single module, it assumes the same base architecture and thus may not generalize to models with differing structures.

Switch-based Fusion: LoRA Composite [19] exemplifies a switch-based approach that alternates among multiple LoRA modules during the denoising process, allowing each LoRA’s unique features (e.g., a style or subject) to be integrated at different steps. While effective in many cases, such time-slicing strategies can struggle when finer-grained, dynamic control is needed in sensitive scenarios.

Noise-based Fusion: Methods like NVS-Solver [17] and Time Reversal Fusion [27] integrate multiple generative models by manipulating noise residuals at inference. NVS-Solver adaptively modulates a video diffusion model’s reverse sampling process, inserting scene priors

from given views. Time Reversal Fusion fuses two denoising trajectories (e.g., one forward in time, another reversed) via a weighted average of their intermediate noise estimates. Although these strategies yield promising results, they typically assume that all models share a common architecture and sampling scheme, limiting their versatility in more heterogeneous fusion scenarios.

2.7 Applications in Dance Video Generation

Dance video generation poses unique challenges, particularly when it comes to preserving the subject's identity and expressions across complex, fast-paced motions. In addition to generating realistic body movements, models must also maintain high-fidelity facial details over multiple frames. Consequently, recent methods have explored multi-conditioned diffusion pipelines that incorporate pose, appearance, and identity constraints to improve temporal stability and personalization [16, 26, 57, 58].

Several notable works illustrate these trends. DisCo [57] disentangles motion from subject appearance by leveraging conditional modules for pose, human identity, and background, producing more consistent results when animating new poses. MagicAnimate [26] employs a video diffusion model coupled with an appearance encoder (e.g., DensePose), aiming to stabilize identity over a longer sequence despite complex motion. MagicPose [58] integrates multi-source attention to guide fine-grained appearance, particularly useful for realistic dance and fashion-driven videos. Meanwhile, AnimateAnyone [16] uses a reference image and pose sequence to generate coherent, human-centric animations, though it may still exhibit limitations in consistently preserving facial features under challenging poses.

These approaches underscore the importance of multi-condition inputs, such as pose keypoints and reference images, for controlling both motion and appearance. They also highlight open problems such as flickering, facial drift, and performance overhead when handling high-resolution or long-duration content. In response, model-fusion strategies offer a promising way to combine specialized modules (e.g., motion-centric vs. face-centric) without large-scale re-training, ensuring both motion fidelity and identity consistency in dance video generation.



2.8 Summary of Literature Review

Diffusion models have substantially advanced both image and video generation, offering stable training and high-fidelity results. Techniques like DDPMs and Stable Diffusion streamline large-scale synthesis, while modules such as ControlNet and IP-Adapter provide finer-grained control for style, identity, and spatial guidance. Yet, challenges in video coherence and model fusion persist, particularly when architectures differ. AdaptiveFuse aims to address these gaps by enabling model-agnostic noise fusion for complex generative tasks.





Chapter 3. Character LoRA and Style LoRA Fusion

This chapter introduces a novel noise-based fusion strategy, referred to as **AdaptiveFuse**, that integrates multiple LoRAs, specifically Character LoRA and Style LoRA, within a single diffusion model. By operating at the noise residual level, rather than merging model weights directly, AdaptiveFuse offers a flexible and adaptive framework to maintain both a character’s unique identity and a chosen artistic style. The following sections detail the motivation, methodological approach, experimental setup, and results. These findings highlight how noise-based fusion can surpass traditional methods in achieving both identity consistency and stylistic richness in image generation.

3.1 Motivation

Producing a seamless combination of distinct attributes, such as a character’s identity and a given artistic style, is crucial to generating personalized yet high-quality images. LoRA has emerged as an effective fine-tuning mechanism for pre-trained diffusion models, enabling specialization in particular identities or styles without demanding comprehensive re-training. However, merging multiple LoRAs, each specializing in a different aspect, is non-trivial. Conventional methods often merge LoRA weights directly [18], which can lead to conflicts between specialized adaptations and thus degrade performance.

This work is driven by the necessity to fuse Character LoRA and Style LoRA while preserving both the character’s identity and the desired artistic style. By devising a more sophisticated fusion approach, we seek to enhance both image fidelity and stylistic variety, circumventing the shortcomings of existing fusion paradigms.

3.2 Methodology

3.2.1 AdaptiveFuse Framework

AdaptiveFuse introduces an innovative approach for combining multiple LoRAs via noise residual manipulation during the denoising process. In contrast to traditional approaches that directly merge LoRA weights, AdaptiveFuse operates at the noise level, enabling a model-agnostic and more dynamically adaptable synthesis.

As illustrated in Figure 3.1, our framework features two LoRA-augmented diffusion U-Nets: the **Character LoRA-Net** (top) and the **Style LoRA-Net** (bottom). Both U-Nets receive pose inputs from ControlNet (via OpenPose keypoints) to ensure that the synthesized image respects the intended body posture. Additionally, text embeddings from a text CLIP encoder guide each network’s cross-attention layers, informing the overall visual theme or semantic content.

The Character LoRA-Net incorporates a LoRA module designed to preserve the subject’s identity and key features, injecting low-rank parameter updates into the spatial and cross-attention blocks of the U-Net. In parallel, the Style LoRA-Net applies a LoRA module optimized for rendering a particular artistic style (e.g., oil painting or sketch). Both of these specialized U-Nets produce their own noise residuals at each denoising step.

These residuals are then passed to the **Fusion Module**, which adaptively combines them according to the selected fusion strategy (e.g., half-weight, switch-based, or noise-based). The fused latent is iteratively denoised until the final timestep, after which a VAE decoder, $\mathcal{D}$, converts the latent into the final synthesized image. By segregating identity-related adaptations (top U-Net) from style-specific ones (bottom U-Net), our approach allows for precise control over both subject fidelity and artistic effect, while remaining compatible with the pose constraints specified by ControlNet.

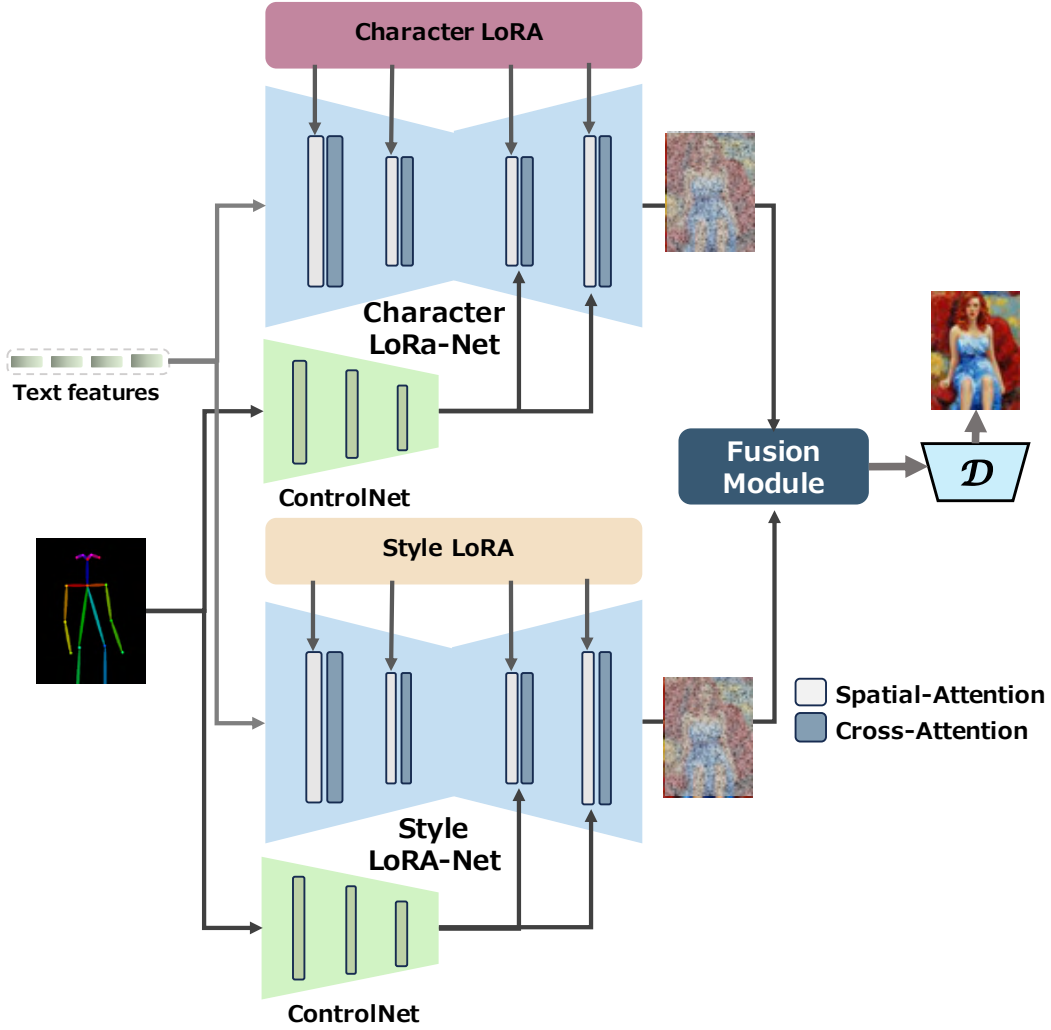


Figure 3.1 High-level overview of the AdaptiveFuse framework for LoRA fusion. For Character LoRA-Net, character LoRA is applied. *ControlNet OpenPose* encodes the pose sequence. Meanwhile, Style LoRA-Net is employed for style feature. The U-Nets of Character LoRA-Net and Style LoRA-Net then compute noise residuals, which are integrated by the Fusion Module. The final denoised image are decoded by the VAE into the generated output.

3.2.2 Noise-based Fusion Strategies

AdaptiveFuse adopts a noise-level fusion philosophy inspired by approaches such as NVS-Solver [17] and Time Reversal Fusion [27], which highlight that noise-space manipulations can unite multiple diffusion models with minimal conflict. Rather than blending network weights, AdaptiveFuse dynamically combines the noise residuals yielded by each LoRA-augmented U-Net.

At each timestep t , the two models predict noise residuals ($L_{\alpha t}$ from Character LoRA-Net,



and $L_{\beta t}$ from Style LoRA-Net). AdaptiveFuse produces a fused latent:

$$L_t = \alpha_t \times L_{\alpha t} + (1 - \alpha_t) \times L_{\beta t}, \quad (3-1)$$

where α_t is an adaptively chosen or user-specified fusion parameter.

3.2.3 Fusion Strategies

To identify an optimal synergy between Character LoRA and Style LoRA, AdaptiveFuse implements multiple strategies:

1. **Switch-based Fusion:** Alternate between LoRAs at predefined intervals (e.g., emphasize Character LoRA in earlier timesteps to establish identity, and transition toward Style LoRA in later steps).
2. **Half-weight Fusion:** Assign equal weights to the LoRA outputs:

$$L_t = 0.5 \times L_{\alpha t} + 0.5 \times L_{\beta t}. \quad (3-2)$$

Though straightforward, this may not always yield an ideal balance.

3. **Noise-based Fusion (Dynamic Adaptation):** Examine noise residual magnitudes or differences at each timestep to adjust α_t :

- *Noise Latents:* Compute a basic noise measure (e.g., sum of absolute values) for each LoRA to guide α_t .

$$\text{noise}_{it} = \sum |\text{latent}_{it}| \quad (3-3)$$

- *Noise Diff:* Assess noise changes over consecutive timesteps to ascertain which LoRA should dominate at a given step.

$$\text{noise}_{it} = \sum |\text{latent}_{it} - \text{latent}_{it-1}| \quad (3-4)$$

- *Noise Ratio*: Track the ratio of noise change over time to continually adapt the fusion ratio.

$$\text{noise}_{it} = \frac{\sum |\text{latent}_{it} - \text{latent}_{it-1}|}{\sum |\text{latent}_{it-1}|} \quad (3-5)$$

4. **Manual Tuning**: Empirically calibrate α_t based on visual evaluations, allowing a human-in-the-loop process.

Figure 3.2 from our original analyses shows how the scheduling of fusion weights evolves over timesteps, showcasing AdaptiveFuse’ s flexible capacity to respect each LoRA’ s specialized domain.

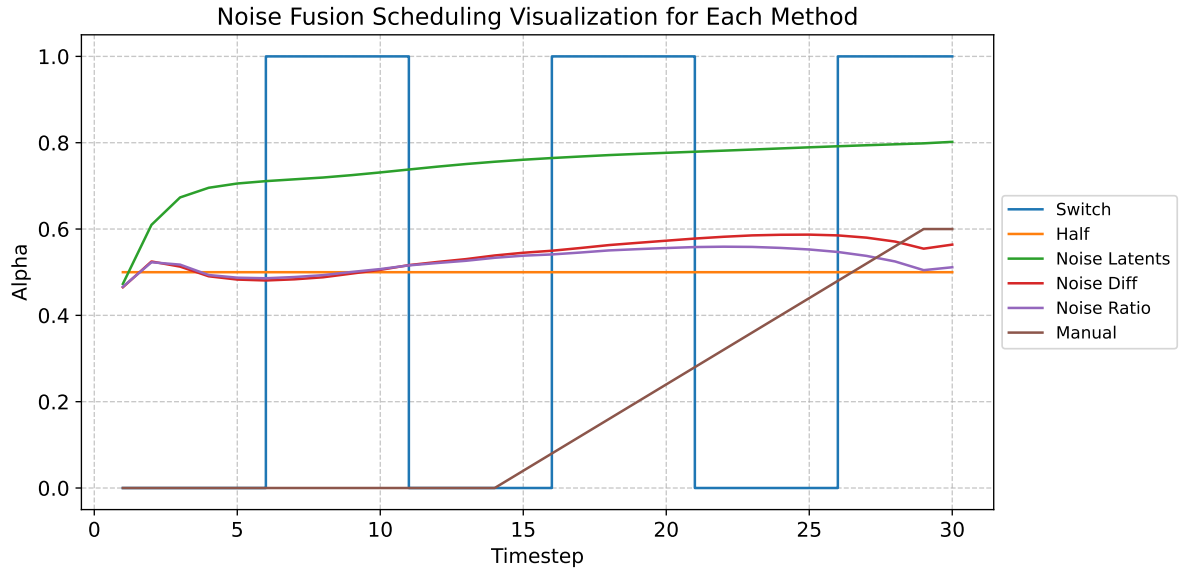


Figure 3.2 Visualization of fusion weights across denoising steps for each scheduling method.

3.2.4 Implementation Details

The fusion process adheres to:

$$L_t = \alpha_t \times L_{\alpha t} + (1 - \alpha_t) \times L_{\beta t},$$

where $L_{\alpha t}$ and $L_{\beta t}$ are noise residuals derived from Character LoRA and Style LoRA, respectively, and α_t manages each LoRA’ s influence at timestep t . By dynamically tuning α_t , using



either noise-based cues or manual adjustments, AdaptiveFuse achieves an effective equilibrium between identity preservation and style fidelity.

Algorithm 1 summarizes the generation steps, showcasing the flexibility of integrating various fusion strategies.

Algorithm 1 AdaptiveFuse Generation Process (Short Version)

Input: Pose $\{P\}$, text prompt T_p

Output: Generated output O

- 1: **Initialize:** Character LoRA-Net, Style LoRA-Net, and latent $\mathbf{z}_T$ from noise
 - 2: **for** $t = T$ **down to** 1 **do**
 - 3: $L_{\alpha t} \leftarrow \text{CharacterLoRA-Net}(\mathbf{z}_t, t, T_p, P)$
 - 4: $L_{\beta t} \leftarrow \text{StyleLoRA-Net}(\mathbf{z}_t, t, T_p, P)$
 - 5: **Compute** α_t (e.g. switch-based, half-weight, etc.)
 - 6: $L_{\text{fused}} \leftarrow \alpha_t \times L_{\alpha t} + (1 - \alpha_t) \times L_{\beta t}$
 - 7: $\mathbf{z}_{t-1} \leftarrow L_{\text{fused}}$
 - 8: **end for**
 - 9: **Decode** $\mathbf{z}_0$ to get output O
 - 10: **return** O
-

3.3 Experimental Setup

3.3.1 Base Models

Our experiments utilize publicly available LoRA models in conjunction with stable diffusion backbones. These LoRA models encompass high-resolution images annotated with character identities and accompanying artistic styles. Such diversity validates the breadth and robustness of the tested fusion strategies.

- **Character Identity:** Employing Scarlett Johansson LoRa weights [65].
- **Artistic Styles:** Employing style LoRAs, such as lineart style [66], oil painting style [67] and ink style[68].

3.3.2 Baselines

We compare our proposed method against traditional LoRA weight merging techniques through simple averaging, as shown in [18].

3.3.3 Implementation Details

The AdaptiveFuse framework is implemented using PyTorch and executed on a single NVIDIA 3090 GPU with 24GB of memory. Generated videos have a resolution of 768×1024 pixels, processed over 30 denoising steps to balance quality and computational efficiency. Pre-trained models, including Character LoRA-Net and Style LoRA-Net, are utilized without further fine-tuning, emphasizing the training-free nature of the fusion approach.

3.3.4 Evaluation

We primarily rely on qualitative evaluations, visual inspections and comparisons, to confirm that the fused outputs preserve character identity and exhibit coherent stylistic adaptations.

3.4 Results

Figure 3.3 illustrates the fusion of three main inputs: a *ControlNet* pose that dictates the subject’s body arrangement, a *Character LoRA* tailored to preserve individual identity (facial structure and other personal traits), and a *Style LoRA* responsible for rendering a distinct artistic look (e.g., ink drawing or oil painting). By combining these elements, we seek images that retain both the subject’s recognizable features and the chosen style, all while conforming to the specified pose.

In each row, four distinct methods are shown, from left to right: *Merge*, *Switch*, *Half*, and *Noise Latents*. The first three approaches highlight common limitations of weight-centric fusion, such as partly losing the subject’s identity or muting certain style aspects. In contrast, the final column (*Noise Latents*) demonstrates how our noise-based integration strategy adaptively balances the two LoRAs’ contributions: identity details (like the character’s facial shape) remain sharply defined, while the stylistic traits (brushstrokes, color schemes) appear more

vivid. As a result, the “Noise Latents” method avoids the abrupt inconsistencies seen in simpler methods and yields more coherent and visually appealing images.



Figure 3.3 Qualitative demonstration of fusing Character LoRA and Style LoRA with ControlNet. Unlike earlier weight-merging approaches, our noise-based method successfully retains identity and integrates style.

Key Observations:

- **Identity Preservation:** The character’s defining features remain consistent, closely mirroring the reference images.
- **Style Integration:** AdaptiveFuse incorporates the intended artistic style without overshadowing the character’s essence.
- **Overall Quality:** Compared to conventional fusion strategies, AdaptiveFuse yields sharper, more detailed results with fewer artifacts.



3.5 Discussion

3.5.1 Global Space Applicability

AdaptiveFuse’ s strong performance stems largely from its noise-level fusion strategy, which focuses on aggregating noise residuals rather than merging network weights. By blending each LoRA’ s specialized output at inference, the method mitigates direct interference between different adaptations (e.g., subject identity vs. artistic style). Although we highlight examples with specific objects or character features, this same principle extends to broader, scene-wide integrations. Manual tuning of the fusion parameter α_t can further tailor the final visual appearance to specific user preferences or domain requirements, suggesting that noise-based fusion can be applied effectively at both local (object-level) and global (scene-level) scales.

3.5.2 Theoretical Basis of AdaptiveFuse

Diffusion models rely on a forward pass that incrementally adds noise to the data, followed by a reverse pass that systematically removes it, eventually reconstructing a clean output. When a LoRA module modifies certain U-Net parameters, it alters the denoising trajectory in a specialized way (e.g., preserving facial identity or injecting stylistic traits). By fusing noise predictions instead of raw parameters, AdaptiveFuse isolates each LoRA’ s unique contribution at every timestep, leveraging the notion that different personalization modules exhibit distinct “strengths” at different stages of the denoising process. Unlike weight-based merging, where specialized modifications can clash, noise-level manipulation keeps each LoRA’ s domain-specific adjustments more intact, thereby enabling a scalable and flexible mechanism to blend multiple model specializations.

3.5.3 Limitations

While our experiments demonstrate that AdaptiveFuse can seamlessly combine LoRAs for style and identity within the Stable Diffusion ecosystem (including ControlNet and LoRA modules), its generality beyond these architectural variants needs further verification. In Chapter 4,



we partially address this by applying AdaptiveFuse to a more complex scenario, like dancing video generation, but additional studies with distinctly different diffusion architectures would better validate its full flexibility.

3.6 Summary

This chapter introduced AdaptiveFuse, a noise-based fusion technique for merging Character LoRA and Style LoRA within a single diffusion model. By focusing on noise-level blending rather than weight-level merging, our approach effectively balances identity fidelity and stylistic expressiveness. Experimental results illustrate sharper, more coherent images and reduced artifacts compared to conventional weight-centric methods. This confirms that noise-space integration can unify diverse LoRAs into a more refined, flexible, and high-quality generative pipeline.



Chapter 4. Fusion for Dancing Video Generation

Building on the success of Character LoRA and Style LoRA Fusion, this chapter presents a novel application of the AdaptiveFuse framework to the challenging task of dancing video generation. By seamlessly integrating models specialized in global motion dynamics with those excelling in preserving fine-grained facial details, AdaptiveFuse addresses long-standing issues in consistency and realism. The following sections detail the motivation, methodology, and experimental setup, as well as evaluate the performance of our approach. Through adaptive noise-based fusion, we demonstrate significant improvements in both temporal coherence and identity preservation, pushing the boundaries of high-quality, personalized dancing video synthesis.

4.1 Motivation

Generating high-quality dancing videos that preserve fine-grained facial details and expressions is a multifaceted challenge. This task necessitates the integration of multiple specialized capabilities to ensure both temporal dynamics and identity consistency. Existing diffusion-based animation methods, while proficient in capturing overall motion and appearance, often falter in maintaining consistent and detailed facial features. Notably, methods like *DisCo* [57] and *MagicAnimate* [26] exhibit inconsistencies and artifacts in facial regions, undermining the realism and identity preservation of the generated videos.

To address these limitations, there is a pressing need for an approach that seamlessly integrates specialized models focused on different aspects of video generation. Specifically, combining a model adept at capturing temporal motion with another specialized in refining facial details can enhance the quality and realism of dancing videos. This fusion aims to take advantage of the strengths inherent in each model, thereby ensuring both dynamic motion and high-fidelity facial features in the final output.

4.2 Methodology

This section introduces the AdaptiveFuse framework, designed to enhance dancing video generation by seamlessly integrating multiple pre-trained diffusion models. The framework leverages specialized models to capture both overall motion dynamics and fine-grained facial details, ensuring high-quality and identity-consistent video outputs.

4.2.1 AdaptiveFuse Framework for Dancing Video Generation

Building upon the success of AdaptiveFuse in image generation, this work extends the framework to the domain of dancing video generation. The approach involves fusing two key components:

- **AnimateAnyone-Net:** A state-of-the-art human image animation framework that excels in capturing overall motion and appearance based on reference images and pose sequences [16].
- **Face-Net:** A specialized module designed to refine facial features and expressions, ensuring high-fidelity and identity-consistent face generation. Face-Net integrates *IP-Adapter FaceID* [25] and *ControlNet* [24] to enhance facial feature generation.

To better understand the complementary strengths of each component, we also visualize the video generation results obtained using only Face-Net and only AnimateAnyone-Net in Figure 4.1. Although Face-Net encounters challenges in generating clothing and backgrounds, it specializes in generating high-quality facial details, capturing the likeness and expressions of the reference face. In contrast, AnimateAnyone-Net focuses on the overall motion and appearance of the character, but may struggle with fine-grained facial features. The complementary strengths of these two networks highlight the importance of employing an adaptive fusion strategy to effectively combine their outputs.



Figure 4.1 Visualization of the video generation results obtained using only Face-Net and AnimateAnyone-Net. Face-Net specializes in generating high-quality facial details, while AnimateAnyone-Net focuses on the overall motion and appearance of the character. The complementary strengths of these networks underscore the importance of adaptive fusion for optimal results.

4.2.2 Fusion Module

The Fusion Module is the core component that integrates the outputs of AnimateAnyone-Net and Face-Net. It employs a mask-based approach to combine the refined facial features

from Face-Net with the body and background generated by AnimateAnyone-Net. The fusion process is governed by the noise residuals computed at each denoising step, ensuring a smooth and coherent integration.

Mathematically, the fusion process is described as:

$$L_t = M \times [\alpha_t \times L_{at} + (1 - \alpha_t) \times L_{ft}] + (1 - M) \times L_{at} \quad (4-1)$$

where:

- L_t is the combined latent at timestep t .
- L_{at} is the latent output from AnimateAnyone-Net.
- L_{ft} is the latent output from Face-Net.
- M is the face mask identifying the facial region.
- α_t is the fusion parameter controlling the influence of each network at timestep t .

4.2.3 Face-Net Integration

Face-Net integrates *IP-Adapter FaceID* [25] and *ControlNet* [24] to enhance facial feature generation. *IP-Adapter FaceID* extracts identity-specific information from the reference image, ensuring that the generated faces maintain the unique characteristics of the subject. *ControlNet* conditions the face generation on desired poses using face masks, enabling dynamic expressiveness and consistency with the intended motion.

This dual approach ensures that the generated faces are both identity-preserving and dynamically expressive, addressing the limitations observed in existing methods where facial details are often inconsistent or distorted.

4.2.4 Noise-based Fusion Schedulers

To address discrepancies in noise schedules between AnimateAnyone-Net and Face-Net, AdaptiveFuse employs various fusion schedulers that adapt the fusion parameters based on noise levels, as we mentioned at 3.2.3 Fusion Strategies.



4.2.5 Algorithmic Overview

Figure 4.2 provides an overview of the AdaptiveFuse framework, illustrating the interaction between AnimateAnyone-Net, Face-Net, and the Fusion Module. The algorithmic process is summarized in Algorithm 2.

Algorithm 2 AdaptiveFuse Dancing Video Generation

Input: Reference image I , pose sequence $\{P_0, \dots, P_{n-1}\}$, face mask M , text prompt T_p

Output: Generated video V

- 1: **Initialize:** AnimateAnyone-Net, Face-Net, and latent $\mathbf{z}_T$ from noise
- 2: **for** $t = T$ **down to** 1 **do**
- 3: $L_{at} \leftarrow \text{AnimateAnyone-Net}(\mathbf{z}_t, t, I, P_t, T_p)$
- 4: $L_{ft} \leftarrow \text{Face-Net}(\mathbf{z}_t, t, I, P_t, T_p)$
- 5: **Compute** α_t (via scheduler or manual)
- 6: **Fuse latents:**

$$L_t = M \times [\alpha_t \times L_{at} + (1 - \alpha_t) \times L_{ft}] + (1 - M) \times L_{at}$$

- 7: $\mathbf{z}_{t-1} \leftarrow L_t$
 - 8: **end for**
 - 9: $V \leftarrow \text{Decode}(\mathbf{z}_0)$
 - 10: **return** V
-

4.3 Experimental Setup

4.3.1 Datasets

To evaluate the robustness and generalizability of our models, we employ two benchmark datasets:

- **TikTok Dataset** [69]: Comprises approximately 350 single-person dance videos, each spanning 10-15 seconds, offering diverse subjects and motions.

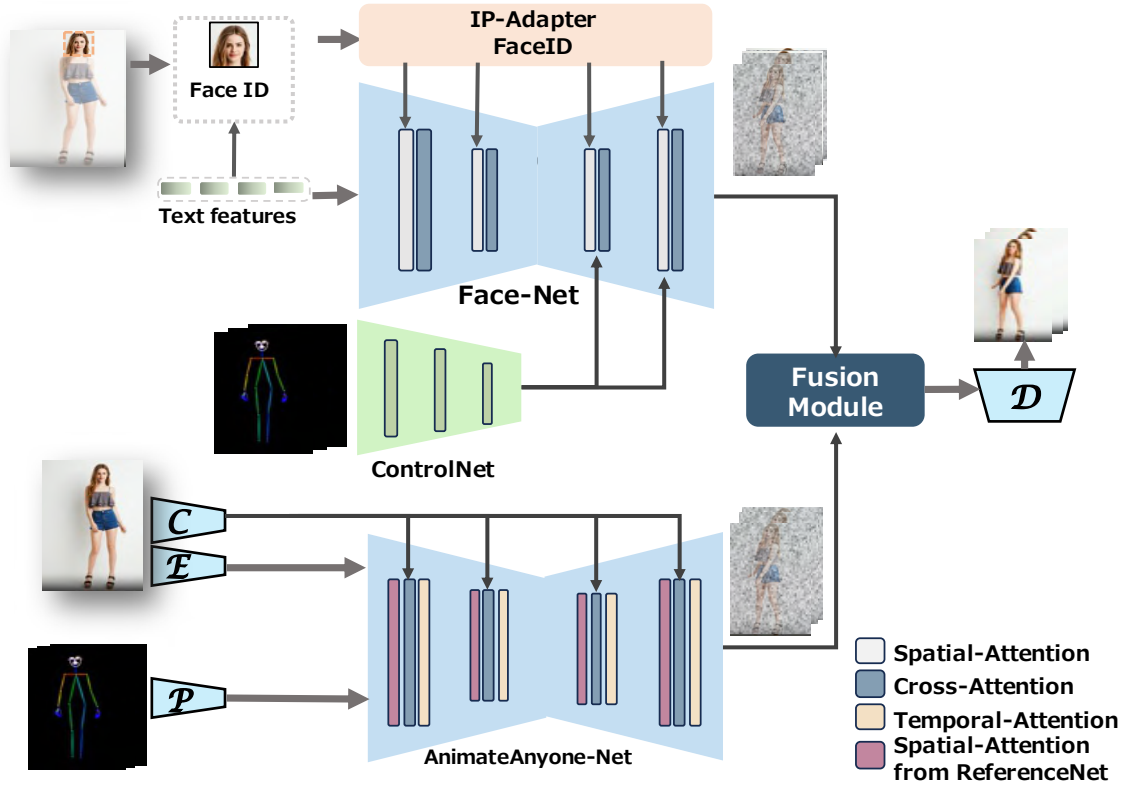


Figure 4.2 Overview of the AdaptiveFuse framework. For Face-Net, the face image is extracted using InsightFace, and *IP-Adapter FaceID* extracts face features. *ControlNet OpenPose* encodes the pose sequence. For AnimateAnyone-Net, the reference image is encoded by the VAE Encoder and CLIP to extract appearance features. The Pose Guider encodes the pose sequence. The encodings are respectively input to AnimateAnyone-Net and Face-Net, which contain U-Nets that iteratively compute noise residuals for the video frames. The Fusion Module uses the noise residuals and fuses them by applying weights with a face mask. The denoised frames are finally decoded by the VAE Decoder into the video clip.

- **UBC Dataset** [70]: Contains a total of 600 videos, depicting various actions such as standing, moving sideways, and turning around, providing a broad range of motion dynamics.

For each dataset, the first frame of each video serves as the reference image, while the remaining frames are extracted to form a pose sequence representing the dynamic movement. Pose estimation is performed using DWPose [71], and facial identity information is extracted using InsightFace [72]. The facer tool [73] is utilized for mask extraction to accurately identify facial regions.

4.3.2 Implementation Details

The AdaptiveFuse framework is implemented using PyTorch and executed on a single NVIDIA 3090 GPU with 24GB of memory. Generated videos have a resolution of 512×512 pixels and consist of 24 frames, processed over 30 denoising steps to balance quality and computational efficiency. Pre-trained models, including AnimateAnyone-Net and Face-Net, are utilized without further fine-tuning, emphasizing the training-free nature of the fusion approach.

4.3.3 Evaluation Metrics

We evaluate the proposed method by employing a combination of quantitative and qualitative metrics:

- **Quantitative Metrics:**

- **L1 Error:** Measures pixel-level discrepancies between generated and reference frames.
- **PSNR (Peak Signal-to-Noise Ratio):** Evaluates image quality based on pixel intensity differences [74].
- **SSIM (Structural Similarity Index):** Assesses the structural similarity between generated and reference frames [75].
- **LPIPS (Learned Perceptual Image Patch Similarity):** Measures perceptual similarity using deep features [76].
- **FID (Fréchet Inception Distance):** Evaluates the distribution alignment between generated and real samples [77].
- **FVD (Fréchet Video Distance):** Assesses the temporal consistency and realism of generated videos [78].
- **Adaface Score:** Computes cosine similarity between features of generated and reference faces to evaluate identity preservation [79].

- **CLIP Score (Image CLIP Score):** Measures alignment between generated videos and corresponding image input [39].
 - **Consistency Score:** Quantifies temporal stability and smoothness of motion across frames [80].
- **Qualitative Evaluation:**
 - Evaluates the realism and coherence of generated videos, focusing on facial detail preservation and motion fidelity.

4.3.4 Baselines

The proposed method is compared against several state-of-the-art baselines:

1. **DisCo** [57]: Integrates disentangled condition modules for pose, human, and background into a pre-trained diffusion model for human image animation.
2. **MagicAnimate** [26]: Utilizes a video diffusion model and an appearance encoder for temporal modeling and identity preservation using reference images and DensePose motion sequences.
3. **MagicPose** [58]: Composes a multi-source attention module providing detailed appearance guidance for human-centric video generation.
4. **AnimateAnyone** [16]: A pre-trained human image animation framework that generates consistent dancing videos from reference images and pose sequences but may lack fine-grained facial detail preservation.

4.4 Results

4.4.1 Quantitative Results

Table 4.1 presents a quantitative comparison of our proposed method with state-of-the-art baselines across various image quality, video quality, and consistency metrics. Our approach



excels in preserving facial identity and generating realistic faces, as evidenced by the highest Adaface score of 0.390, highlighting the effectiveness of our Face-Net module in refining facial features and maintaining consistency with the reference image.

Regarding image quality, our method obtains the highest SSIM of 0.805 and the lowest L1 error of $4.94e-05$, indicating excellent structural similarity and pixel-level accuracy. We also achieve a PSNR of 19.4, matching the top-performing AnimateAnyone. The LPIPS score of 0.153 is comparable to AnimateAnyone’s 0.150, suggesting similar perceptual quality.

For video quality, our method exhibits the lowest FVD of 13.1, slightly surpassing AnimateAnyone’s 13.2, demonstrating improved temporal coherence and visual quality of the generated videos. The FID score of 24.7 is the second lowest, closely following AnimateAnyone’s 23.3, indicating the realistic nature of the synthesized video frames.

Regarding consistency and faithfulness to the input, our method achieves the highest CLIP score of 0.900, showcasing a strong alignment between the generated videos and their corresponding text descriptions. Additionally, our consistency score of 0.952 is on par with AnimateAnyone’s 0.953, reflecting the temporal stability of the generated videos.

Notably, our approach significantly outperforms other baselines, including DisCo, MagicAnimate, and MagicPose, in terms of facial identity preservation and overall face quality. DisCo and MagicAnimate exhibit lower Adaface scores of 0.115 and 0.274, respectively, indicating their limitations in generating consistent and realistic faces. MagicPose achieves a higher Adaface score of 0.375 but still falls short of our method’s performance.

The superior Adaface score achieved by our method underscores the effectiveness of the proposed Face-Net module and the adaptive fusion technique. By dedicating a specialized network to refine facial features and seamlessly integrating its output with the content generated by AnimateAnyone, our approach ensures high-quality and identity-consistent face generation in the resulting videos.

Overall, the quantitative evaluation demonstrates the state-of-the-art performance of our method in generating realistic and consistent human dance videos, with a particular emphasis on enhancing facial quality and preserving identity. The combination of the Face-Net module and the adaptive fusion strategy enables our approach to outperform existing methods in the critical

aspect of face generation while maintaining competitive results in other evaluation metrics.

Table 4.1 Quantitative comparisons of our method with recent SOTA methods. $\uparrow$ indicates higher is better, and $\downarrow$ indicates lower is better. **Bold indicates the best result in each metric.**

Metric	DisCo	MagicAnimate	MagicPose	AnimateAnyone	Our Method
PSNR $\uparrow$	16.6	7.86	13.7	19.4	19.4
SSIM $\uparrow$	0.657	0.524	0.689	0.797	0.805
L1 $\downarrow$	7.75e-05	33.1e-04	16.2e-04	5.06e-05	4.94e-05
LPIPS $\downarrow$	0.235	0.457	0.323	0.150	0.153
FID $\downarrow$	57.4	62.2	37.1	23.3	24.7
FVD $\downarrow$	53.4	83.6	71.5	13.2	13.1
Consistency $\uparrow$	0.912	0.943	0.924	0.953	0.952
CLIP $\uparrow$	0.831	0.829	0.854	0.899	0.900
Adaface $\uparrow$	0.115	0.274	0.375	0.354	0.390

4.4.2 Qualitative Analysis

Figure 4.3 presents a side-by-side comparison of dancing video frames generated by our method (*rightmost column*) versus four baseline methods (DisCo, MagicAnimate, MagicPose, and AnimateAnyone). Each row corresponds to a different *reference image* (leftmost photos) and *pose* (OpenPose keypoints shown in the second column from the left). The reference image determines the subject’s identity and appearance, while the reference pose constrains the target motion to be replicated.

Columns under each baseline illustrate how well the generated frames preserve both global motion (matching the reference pose) and local features (notably the face region). Specifically, we highlight two primary aspects:

- **Facial Fidelity.** Observe that in DisCo’s results, the subject’s face sometimes appears distorted or mismatched, and MagicAnimate occasionally loses sharp facial features. MagicPose shows improved facial alignment but can still produce artifacts (e.g., slight blurring). AnimateAnyone generally maintains consistent head poses, but certain facial details are lost or blurred over time.
- **Temporal Coherence.** Our method (rightmost column) shows more stable facial expressions across consecutive frames and fewer flickering artifacts. Even as the subject changes pose, our approach better preserves the subject’s identity (eyes, mouth, and facial structure) and maintains consistent clothing detail over time.

Overall, the proposed method excels at preserving the subject’s identity in the face region while accurately transferring the pose from the reference skeleton, leading to smoother and more realistic dancing videos.

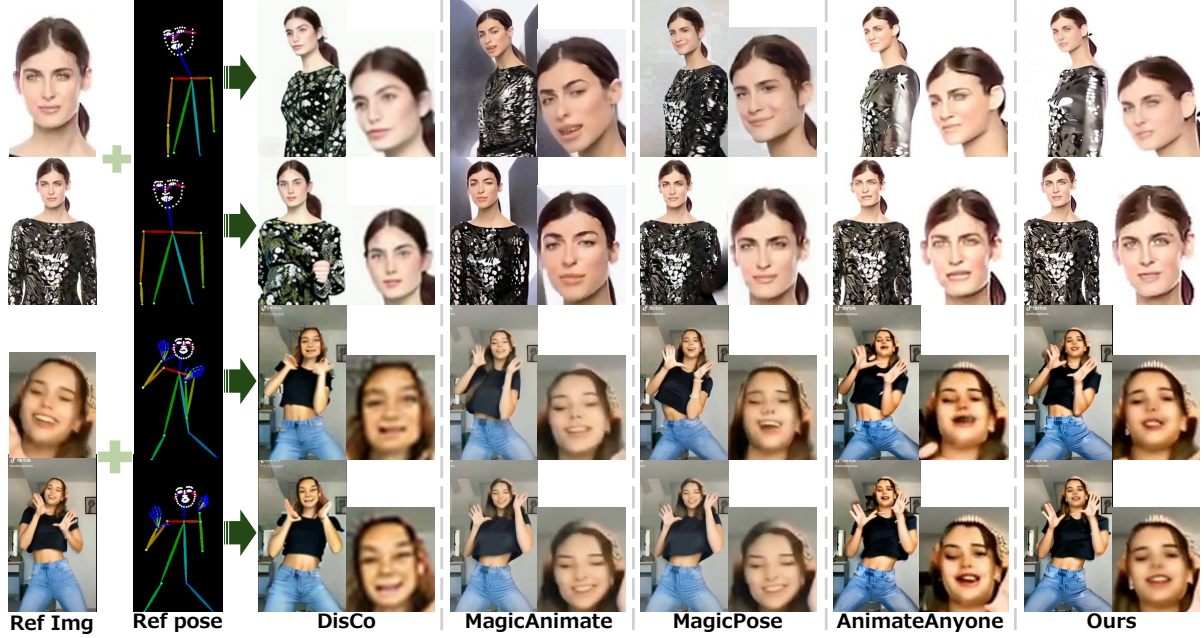


Figure 4.3 Comparative snapshots from videos generated using our method and other baseline methods, with a focus on the face region, demonstrating superior facial detail and temporal coherence in our approach.

Additional Qualitative Comparisons (UBC and TikTok).

Figures 4.4 and 4.5 showcase side-by-side video generation temporal snapshots for reference images from the UBC and TikTok datasets, respectively. Each figure is arranged in rows for different methods (DisCo, MagicAnimate, MagicPose, AnimateAnyone, and ours), highlighting the reference subject, the OpenPose-based *pose* sequence, and the resulting synthesized frames.

- **Reference Image and Pose:** The top rows display the subject’s identity (face, outfit) and the target body movements extracted via pose estimation.
- **Baselines:** DisCo, MagicAnimate, MagicPose, and AnimateAnyone each generate dancing sequences aligned with the reference pose but often introduce blurred facial features or frame-to-frame flickers.

- **Ours:** Our method (bottom rows) preserves sharper facial details and achieves smoother transitions, effectively retaining both identity and motion consistency. Notably, facial expressions remain more stable over time, greatly enhancing realism for dancing videos.

Overall, the comparisons confirm that our approach outperforms baselines on both datasets, delivering superior facial fidelity and temporal coherence in dynamic motion settings.

4.4.3 Ablation Study

To investigate the contribution of various fusion parameters, we conduct a thorough ablation study by comparing switch-based, half-weight, noise-based, and manual tuning strategies against a baseline without fusion. We focus on how well each strategy preserves facial details and overall identity, especially under challenging motion or pose transformations. Table 4.2 and Figures 4.6 and 4.7 present both the quantitative outcomes (e.g., PSNR, SSIM, Adaface) and qualitative snapshots.

Table 4.2 Quantitative ablation study (transposed). $\uparrow$ indicates higher is better, and $\downarrow$ indicates lower is better. Bold indicates the best result in each metric.

Metric	Without Fusing	Switch	Half	Noise Latents	Noise Diff	Noise Ratio	Manual
PSNR $\uparrow$	19.4	19.5	19.5	19.4	19.5	19.5	19.4
SSIM $\uparrow$	0.797	0.805	0.805	0.801	0.805	0.804	0.805
L1 $\downarrow$	5.06e-05	4.94e-05	4.93e-05	5.00e-05	4.94e-05	4.94e-05	4.94e-05
LPIPS $\downarrow$	0.150	0.152	0.151	0.153	0.152	0.152	0.153
FID $\downarrow$	23.3	24.5	24.5	24.8	24.6	24.6	24.7
FVD $\downarrow$	13.2	13.2	13.0	13.1	13.0	13.1	13.1
Consistency $\uparrow$	0.953	0.951	0.952	0.952	0.952	0.952	0.952
CLIP $\uparrow$	0.899	0.898	0.897	0.898	0.898	0.898	0.900
Adaface $\uparrow$	0.355	0.355	0.378	0.371	0.380	0.387	0.390

4.4.3.1 Quantitative Analysis

Table 4.2 presents the metrics for each strategy. Removing the fusion module (*Without Fusing*) decreases the Adaface score from 0.390 to 0.355, confirming the importance of fusion for high-fidelity face generation. **Manual Tuning** attains the highest Adaface score (0.390), underscoring the effectiveness of human guidance. Noise-based approaches (e.g., **Noise Diff**, **Noise Ratio**) also exceed the baseline but remain slightly below **Manual** and **Noise Latents**.

Overall, these ablation results demonstrate that noise-level integration significantly enhances identity consistency and visual quality, with both user-controlled (*Manual*) and automated (*Noise Latents*) schedules excelling in stabilizing facial features across diverse poses.

Qualitative Comparison.

Figure 4.6 provides a close-up view of facial regions across different fusion approaches. Each column corresponds to a specific method:

- **Reference:** Depicts the original subject or target pose for comparison.
- **Without Fusing:** Uses a single network without any noise-level merging, often resulting in distorted or inconsistent faces.
- **Switch, Half, Noise Latents, Noise Diff, Noise Ratio:** Automated fusion schemes that blend noise residuals at each denoising step. “Switch” and “Half” occasionally suffer from blur or partial identity loss, whereas *Noise Latents*, *Noise Diff*, and *Noise Ratio* generally produce more coherent, stable faces.
- **Manual:** A human-in-the-loop strategy that fine-tunes the fusion parameter α_t based on visual inspection, often leading to the most realistic and expressive facial outcomes.

Visually, **Manual** and **Noise Latents** best preserve both facial detail and identity. In contrast, “Without Fusing,” “Switch,” and “Half” can exhibit more artifacts or blur, highlighting how noise-based fusion (with possible human oversight) is effective in generating clear, consistent facial features.

Figure 4.7 illustrates a similar set of fusion approaches but focuses on varying pose inputs. Columns represent distinct reference poses, while rows correspond to fusion methods (Without Fusing, Switch, Half, Noise Latents, Manual, etc.). We observe:

- **Without Fusing:** Produces inconsistent or artifact-laden facial renderings.
- **Switch, Half:** Relatively simple schedules that mitigate some distortions but risk blur or partial identity mismatch.



- **Noise Latents, Manual:** Offer sharper facial details and more stable expressions across frames, preserving identity more faithfully under different poses.

Hence, both *Noise Latents* and *Manual* emerge as standout strategies for balancing realistic appearance with strong subject fidelity.

4.5 Discussion

The proposed **AdaptiveFuse** framework demonstrates that noise-level fusion can effectively combine multiple specialized diffusion models, such as a motion-centric network (AnimateAnyone) and a face-refinement network (Face-Net), for dancing video generation. In doing so, it offers higher-quality outputs than previous methods in terms of both fidelity and consistency. Nevertheless, certain aspects of the approach warrant closer examination.

First, although the method excels at synthesizing clear and temporally coherent facial details in dancing videos, the computational overhead remains nontrivial. By employing multiple pre-trained models simultaneously, the system runs two specialized U-Nets in parallel, increasing the memory footprint and prolonging inference times. While our empirical results show the performance benefits of leveraging complementary modules, resource-constrained environments may require additional strategies such as parameter sharing, partial inference, or dynamic loading of specific model components.

Second, our exploration of the fusion parameter α_t has centered on either manual “human-in-the-loop” adjustments or heuristic noise-based schedules. Although these strategies are relatively easy to implement and validate, they may not adapt optimally to diverse motion scenarios or complex facial expressions. Future work might incorporate more automated scheduling via reinforcement learning, gradient-based optimization, or adaptive heuristics that consider perceptual metrics and user-driven constraints. Such refinements could help fine-tune the relative weighting of motion and facial-detail models over time, leading to a more seamless blending of specialized capabilities.

Third, our empirical evaluation has focused on single-person dance sequences under controlled conditions. Scaling this approach to more challenging settings, such as multi-person



choreography, highly cluttered scenes, or non-human entities, remains an open research avenue. For instance, modeling interactions among multiple dancers or handling occlusions in a dynamic group context may demand further architectural adaptations or enhanced fusion mechanisms. Likewise, tasks like object-centric video generation, full-scene stylization, and viewpoint transitions may also benefit from noise-level fusion, provided that future research validates the generality of this framework across broader tasks and data domains.

Finally, while our results highlight robust performance even under moderate complexities of motion and expression, situations involving extreme poses, rapid viewpoint changes, or severe lighting variations have not been exhaustively tested. In highly challenging scenarios, more advanced domain-specific priors or scene-level constraints may be necessary to preserve fidelity and ensure temporal coherence. Ongoing improvements to scheduling schemes, domain coverage, and model resiliency, alongside a deeper integration of real-time feedback signals, could strengthen the applicability of AdaptiveFuse and permit a wider range of creative and industrial uses.

Overall, the AdaptiveFuse approach offers a compelling demonstration of how noise-based model fusion can unify specialized strengths within a single pipeline, particularly for the demanding application of dancing video generation. By integrating a face-focused network with a motion-centric model, our strategy attains superior facial fidelity and temporal smoothness, pushing the envelope in identity-preserving human-video synthesis. The next chapter expands on the broader implications of these findings and outlines potential directions for advancing noise-level fusion in future research.

4.6 Summary

In this chapter, we extended the AdaptiveFuse framework to tackle the challenge of dancing video generation, focusing on two primary goals: capturing natural, fluid motion and preserving high-fidelity facial details. By fusing a motion-oriented model (AnimateAnyone-Net) with a face refinement module (Face-Net) at the noise-residual level, our approach achieves improved temporal coherence, more realistic facial features, and stronger identity consistency compared



to existing methods.

A series of quantitative evaluations on benchmark datasets (TikTok and UBC) confirms that our method excels in preserving subject identity (Adaface Score) while maintaining video quality metrics (FVD, SSIM, PSNR) at or above the performance of baselines. Qualitative comparisons further underscore the effectiveness of this strategy: unlike conventional weight-centric or switch-based fusion, noise-level fusion enables each specialized model to contribute its strengths without mutually interfering. An ablation study demonstrates that both user-driven (Manual) and automated (Noise Latents) scheduling strategies enhance face fidelity, validating the flexibility and robustness of noise-based integration. These findings highlight the adaptability of AdaptiveFuse for demanding generative tasks in video domains and illustrate how noise-level fusion enables multiple specialized capabilities, motion accuracy and facial identity, such as to coexist within a single pipeline, paving the way for future advancements in dancing video generation and beyond.

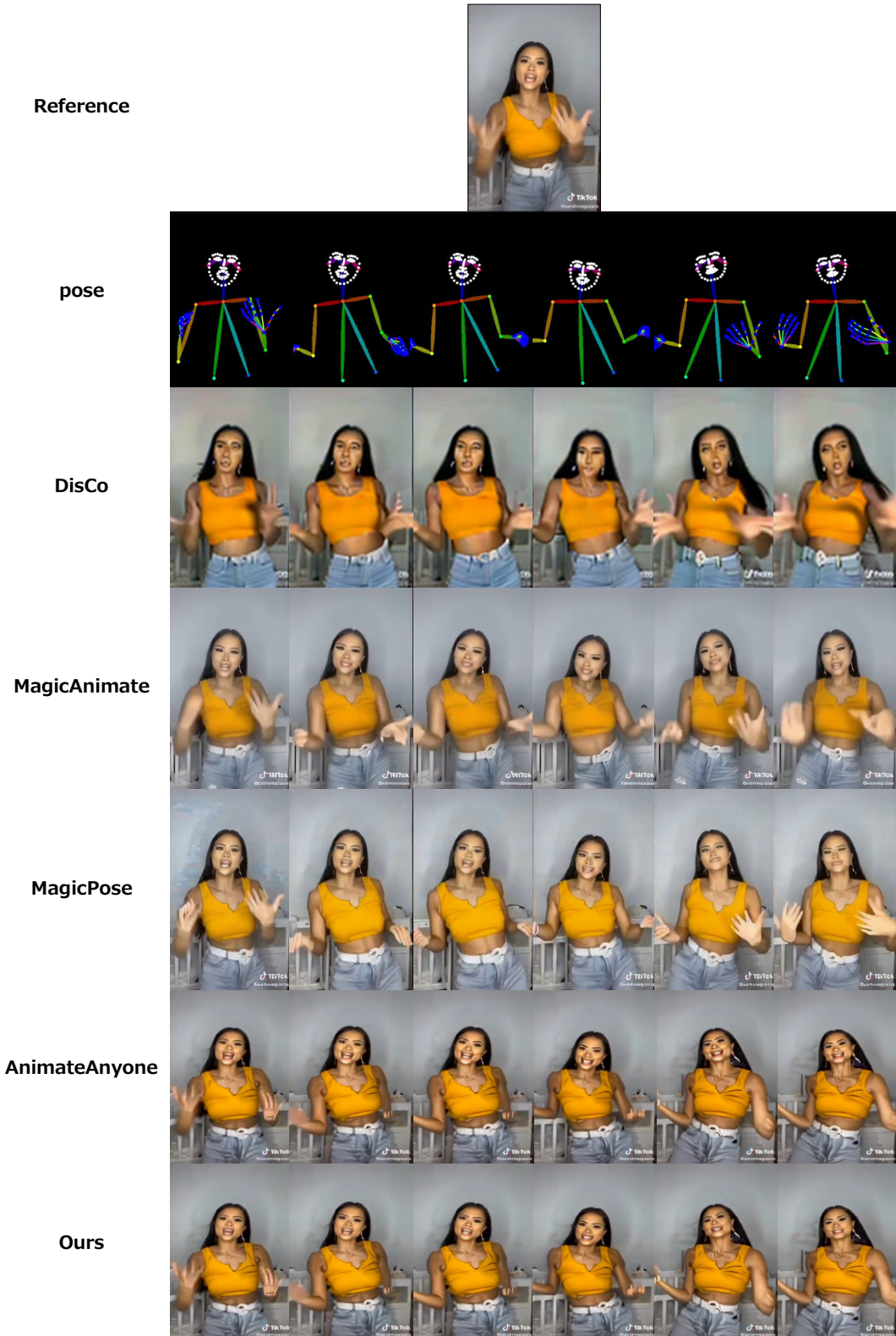


Figure 4.4 Qualitative comparison of video generation results on a reference image from the UBC dataset. We present snapshots from videos generated by our method and state-of-the-art baselines. Our approach demonstrates superior facial detail preservation and temporal coherence, particularly evident in the consistent facial expressions and smooth transitions between frames.

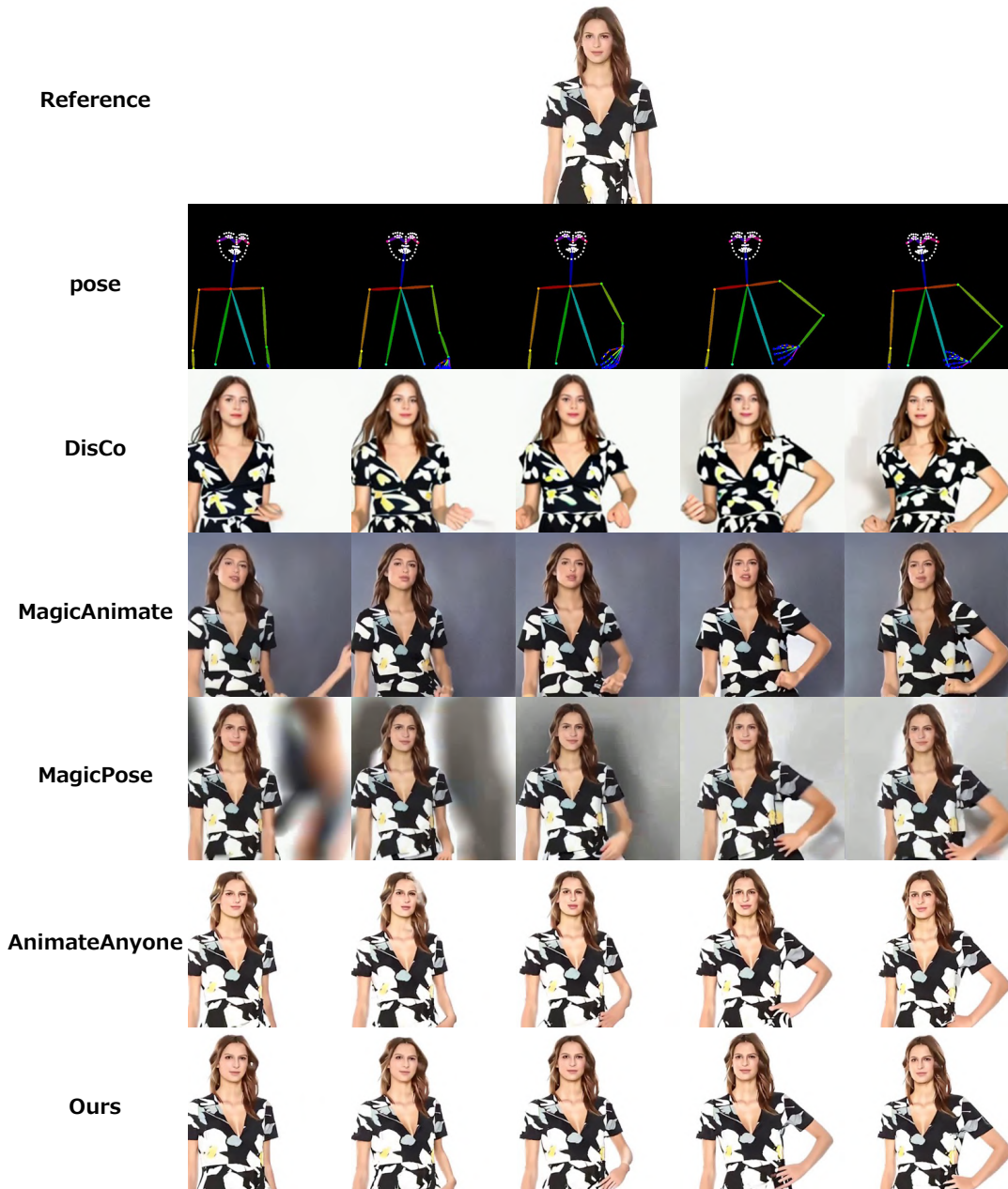


Figure 4.5 Qualitative comparison of video generation results on a reference image from the TikTok dataset. The figure illustrates frame sequences from videos generated by our method and competing approaches. Our method exhibits enhanced facial detail retention and improved temporal stability, evidenced by the consistent facial features and fluid motion across consecutive frames.



Figure 4.6 Qualitative ablation study. We compare the facial regions from videos generated by different fusion parameter settings and the baseline without fusion. Our proposed fusion methods consistently yield superior facial details, preserving subject identity and expressiveness. Notably, the 'Manual' and 'Noise Latents' settings demonstrate particular efficacy in generating realistic and visually appealing facial features, while the 'Without Fusing' baseline exhibits inconsistencies, and the 'Switch' and 'Half' methods show a tendency towards blur.

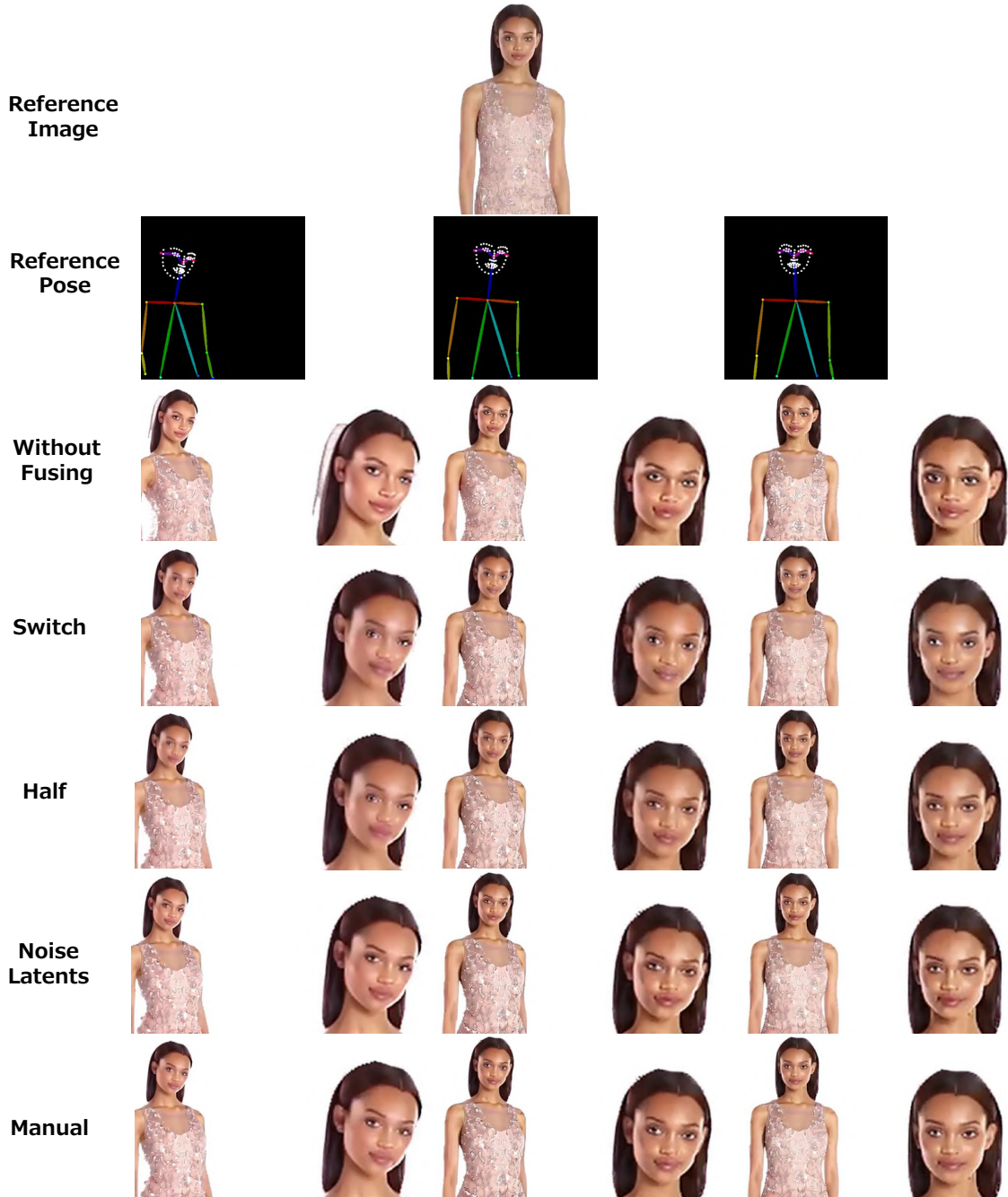


Figure 4.7 Ablation study visualizing the impact of different fusion parameter settings on facial region generation. We compare results from various fusion strategies against a baseline without fusion. Our proposed fusion methods consistently yield superior facial details, preserving subject identity and expressiveness. Notably, the 'Manual' and 'Noise Latents' settings demonstrate particular efficacy in generating realistic and visually appealing facial features, while the 'Without Fusing' baseline exhibits inconsistencies, and the 'Switch' and 'Half' methods show a tendency towards blur.



Chapter 5. Conclusion and Future Work

This chapter concludes by summarizing the main outcomes of this thesis and highlighting directions for further exploration of the AdaptiveFuse framework.

5.1 Conclusion

This thesis aimed to address the difficulties of integrating multiple diffusion models, each specialized in distinct tasks such as artistic style rendering or subject-specific identity, into a single pipeline *without* requiring additional training. Although diffusion models excel at diverse tasks, their reliance on uniform architectures and training paradigms has often posed substantial barriers to combining heterogeneous models. To overcome these obstacles, we introduced AdaptiveFuse, a noise-level fusion framework designed to operate on each model’s predicted noise residuals. By working at the noise level, AdaptiveFuse accommodates architectural variations, distinct training protocols, and differing sampling schemes with minimal friction.

By leveraging the complementary strengths of specialized diffusion models, AdaptiveFuse makes several key contributions to generative modeling:

- **Model-Agnostic Fusion.** It eliminates the need to re-train or restructure large diffusion networks, enabling the seamless combination of models focused on different competencies, from identity preservation to motion generation.
- **Noise-Fusion Principle.** By fusing noise residuals at each denoising step, our framework opens new possibilities for tasks that demand multiple capabilities, such as generating realistic facial details in dancing video sequences.
- **Empirical Validation.** Experiments demonstrate that specialized sub-networks, when combined at the noise level, achieve higher visual fidelity and more coherent results than conventional weight-merging approaches.

Despite these achievements, certain issues remain open for further work. Enhanced scheduling strategies might refine or automate sub-network blending, while partial inference or param-



eter sharing could address the increased resource usage incurred by multiple U-Nets. Testing AdaptiveFuse in more extreme conditions, such as scenes with multiple interacting subjects, abrupt lighting changes, or real-time constraints, would also clarify its robustness and scalability.

In summary, AdaptiveFuse contributes an essential layer of flexibility to diffusion-based generative modeling. By disentangling the fusion process from weight-level constraints and unifying disparate models through noise-space integration, our approach paves the way for advanced applications in video editing, inpainting, and multimodal content creation, unlocking greater potential for generative models to excel in complex, real-world scenarios.

5.2 Future Work

Although AdaptiveFuse successfully addresses multi-objective generation across diverse tasks, several directions remain open for further exploration. In particular, the discussions in earlier chapters suggest that advancing efficiency, refining scheduling, expanding application domains, and improving robustness under challenging conditions can all elevate the impact of noise-fusion techniques. Below, we summarize four key directions for future investigations:

1. Efficiency and Memory Constraints. While combining multiple specialized U-Nets broadens the functional scope of the pipeline, it also increases memory usage and inference time. Pursuing partial-inference strategies, shared parameters, or model compression may help mitigate the added resource demands, enabling wider deployment across devices with limited computational capacity.

2. Enhanced Scheduling Algorithms. Current scheduling methods rely on heuristics or manual interventions for deciding how to fuse noise residuals. A more data-driven or adaptive approach, such as reinforcement learning or gradient-based optimization, could automatically balance the influence of sub-models over time. Such an automated scheduler would not only streamline practical usage but also potentially yield more consistent results in dynamic scenarios, such as human motion.

3. Broader Application Domains. Although this thesis highlighted dancing video gen-



eration and stylized image synthesis, the same noise-fusion paradigm may benefit tasks like multi-person interactions, complex scene modifications, and novel view synthesis. Investigating how AdaptiveFuse performs on challenging domains, such as large-scale 3D environments, real-time video editing, or cross-modal generation, represents a natural next step in expanding its utility.

4. Robustness to Challenging Conditions. Future work should further test AdaptiveFuse under conditions involving occlusions, severe lighting variations, abrupt camera movement, and multi-person scenes. Introducing domain-specific priors or integrative scene models could bolster the framework's resilience to unpredictable environments, thereby fostering more reliable real-world deployments.

By pursuing these directions, subsequent research can amplify AdaptiveFuse's flexibility, computational efficiency, and overall robustness, ensuring that noise-level fusion stands as a robust and widely applicable technique in the rapidly evolving landscape of diffusion-based generative modeling.





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